TH series/TS2000/TS2100

Industrial Robot

MAINTENANCE MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.7
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NUMAZU, JAPAN

Shibaura Machine

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PREFACE

This manual describes how to maintain the SHIBAURA MACHINE's TH series robot and controller. Maintenance and inspection are essential to keep the robot performance for a long time, to prevent failures and to assure safety. Be sure to look through this manual and set up a maintenance program before actually starting the robot.

This manual is composed of the following sections.

Section 1: Maintenance Items
This section describes the maintenance schedule of the robot and controller and items required for inspection and maintenance.

Section 2: Maintenance of the Robot
This section refers to the robot structure and items required for inspection and maintenance of the robot.

Section 3: Maintenance of Controller
This section refers to the robot controller structure and items required for inspection and maintenance of the controller.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Be sure to read through the safety instruction manual provided separately before starting actual work to assure the safe work covering from the robot installation to operation.

[Maintenance and inspection]

Be sure to observe the following items to use the robot safely.

DANGER	
 Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
 Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

CAUTION	
 Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

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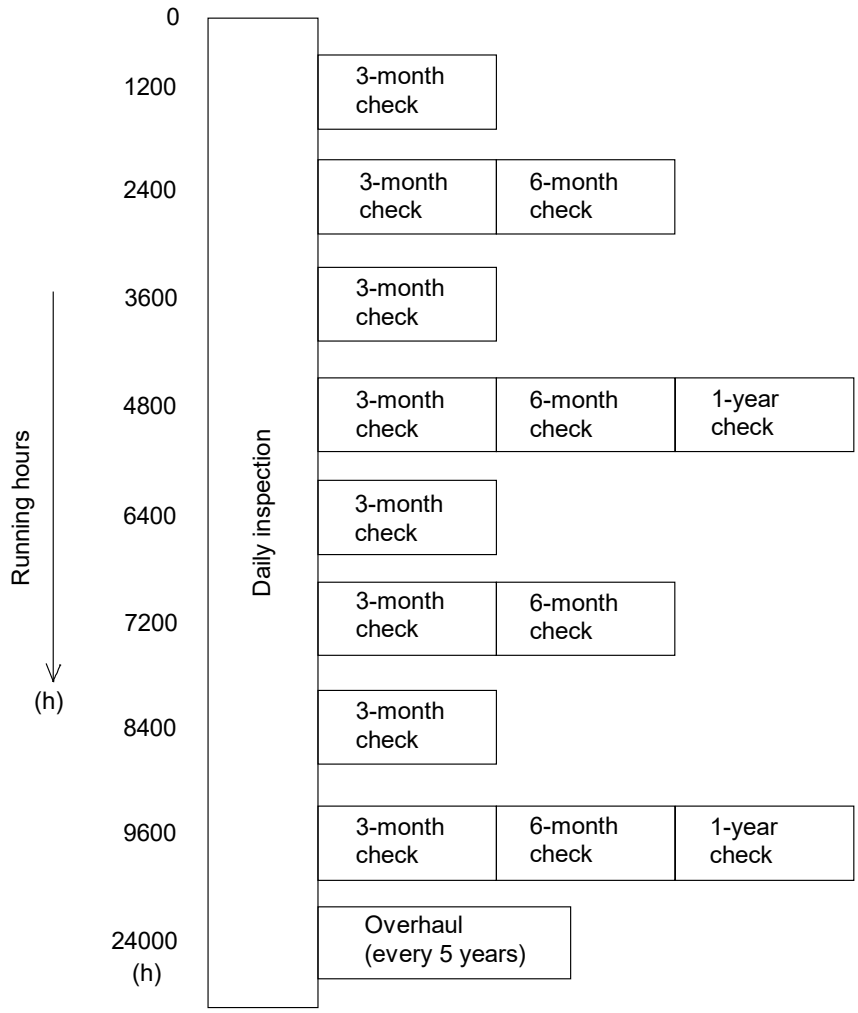
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Section 1 Maintenance Items

1.1 Maintenance Schedule

Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule

- When the system is operated for 16 hours per day:
  16 hours × 25 days × 3 months = 1,200 hours
- When the system is operated for 24 hours per day:
  24 hours × 25 days × 3 months = 1,800 hours

1.2 Items for Maintenance and Inspection

The maintenance comes in the two (2) types; daily inspection executed every day and regular maintenance and inspection executed after the lapse of predetermined duration.
This section describes the items for maintenance and inspection. For the executing procedures, see the relevant paragraph listed in the table below.

a) Daily inspection

No.	Item	Inspection procedures	Relevant Para.
1	Cable ducts (TH450/TH550)	(1) Make sure that any cable duct will not twist, bend or crack locally at the time of robot motion. (2) Make sure that all connectors of the cable ducts are tightened completely.	2.3.1
2	Cables (TH650/TH850 /TH1050)	(1) The cables pass from the robot base to the arm 1 and arm 2. As they pass through the robot interior, remove the arm 2 cover and arm 1 cover to check for an abnormality. Visually check to see that each cable is not locally twisted or bent and that each nylon tube is not locally worn out or cracked.	2.3.2
3	Hand set flange	Make sure that the hand set flange bolts are tightened completely.	2.3.3
	Robot installation bolts	Make sure that the robot base installation bolts are tightened completely.	
4	Vibration and noise	Make sure that each axis can move smoothly.	
5	Change in positioning accuracy	Make sure that the robot can stop at a predetermined position accurately.	
6	Air leakage (When air piping is used)	Make sure that the air will not leak from the robot interior or air joints.	2.3.4

No.	Item	Inspection procedures	Relevant Para.
7	Controller vent holes	(1) Make sure that no obstacle stands in front of the vent holes. (2) Make sure that no contaminant is left in the vent hole.	3.3.1
8	Function of safety devices	Make sure that each safety device connected to the emergency stop pushbutton switch and external operation input signal can function properly.	3.3.2

b) Quarterly (every three (3)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Lubricated state of tool shaft	Make sure that the tool shaft is greased completely and that no rust is developed on it.	2.3.5(a)

c) Semi-annual (every six (6)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Greasing of tool shaft	Apply the grease to the tool shaft.	2.3.5(b)
2	Each axis drive system	Move each axis by hand and make sure that it moves smoothly.	2.3.6
3	Timing belts	(1) Make sure that each timing belt is not worn off or cracked. (2) Check each timing belt for tension.	2.3.7

d) Annual (every year) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Greasing of tool shaft Greasing of gear	Apply the grease to the tool shaft. Fill grease through the grease nipples.	2.3.5(b) 2.3.8
2	Wiring and piping	Make sure that all cables and air tubes are in the good state.	
3	Replacement of position detector batteries (Replace the batteries every year.)	Replace the position detector backup batteries (housed in the cable connector set panel of the robot base unit). If power off time is too long, the battery performance to be weak. So if you no use the robot for a long time (For example: over two months), please replace the batteries when you turn on the robot. If the batteries are still not replaced, a battery alarm occurs for the short term.	2.3.9

e) Overhaul (every five (5) years)

No.	Item	Inspection procedures	Relevant Para.
1	Inspection of consumable parts	Inspect the flexible cables, reduction gears, bearings, tool shaft, timing belts and motors and replace them when necessary.	
2	Battery replacement (Replace the battery every five (5) years.)	Replace the RAM (controller) backup battery. If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage drops, the battery alarm lamp illuminates. Then replace the battery immediately. Alarm message: "MAIN Battery alarm"	3.3.3
3	Replacement of switching power supply unit (Replace the battery every five (5) years.)	As the switching power supply unit uses an aluminum electrolytic condenser, its life is limited. Replace the switching power supply unit (DC5 V, DC24 V) for preventive maintenance. The life of the switching power supply unit differs with load current, running hours and ambient temperature. If the load current is large, the running time is long. If the ambient temperature is high, the life becomes short.	3.3.4

1.3 Maintenance Tools

We recommend to use the following tools for maintaining the robot and controller.

- Phillips screwdrivers
- Hexagonal wrench set: M3 ~ M16
- Assembly wrench set
- Water pump pliers
- Nippers
- Grease guns

1.4 Maintenance Contract and Repair

1.4.1 Maintenance Contract

With the exception of greasing and lubrication, we recommend that the user conclude with us an after-sale service contract on the inspection items for maintenance and inspection to be done semi-annually and after.

1.4.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller.

- Robot model
- Manufacture number
- Date of manufacture
- Operating time

1.4.3 Modification

This robot and controller MUST NOT be modified without the consent from SHIBAURA MACHINE.

! CAUTION

- The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.

Section 2 Maintenance of the Robot

2.1 Cautions on Maintenance and Inspection

When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

2.2 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 2.1 which refers to the TH650/TH850/TH1050 series.

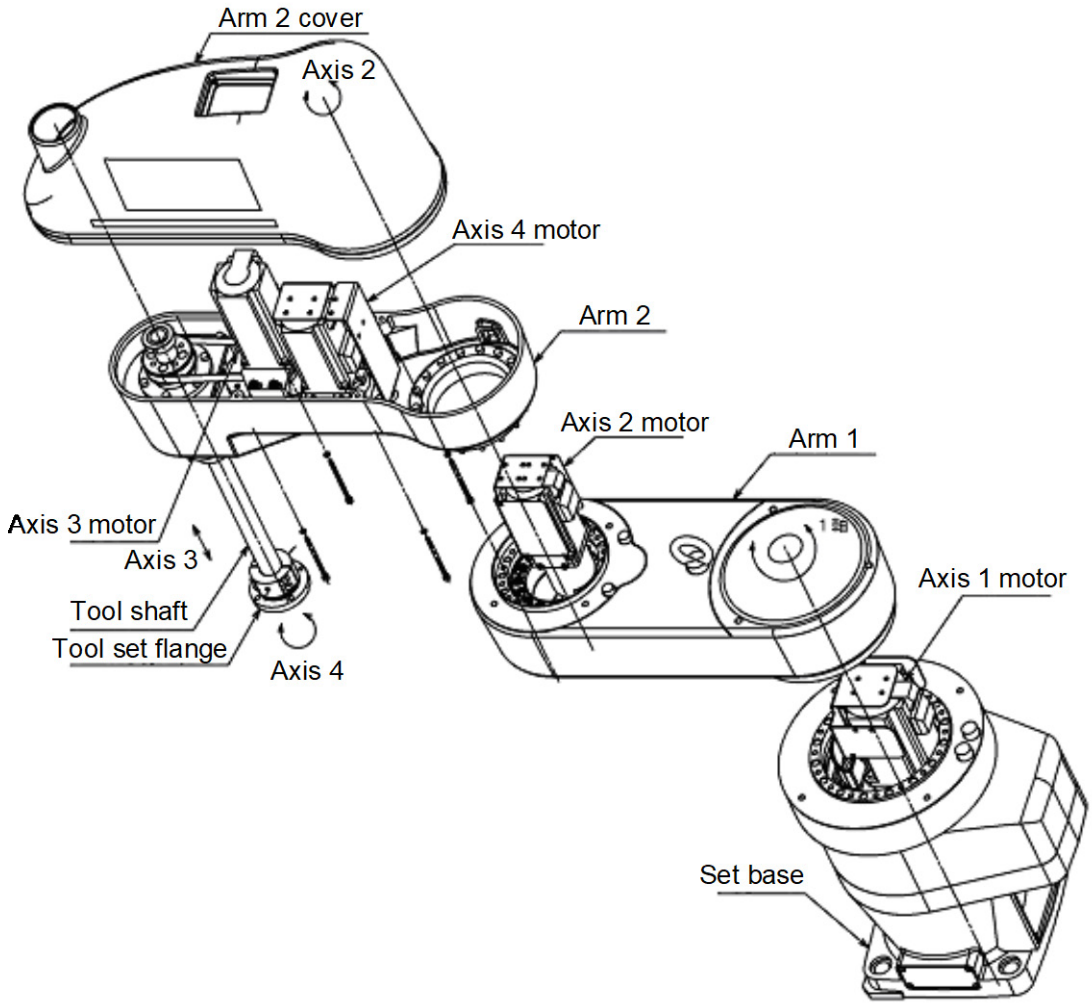


Fig. 2.1 Layout of robot mechanical components of TH650/TH850/TH1050 series

The drive mechanism of each axis of the TH650/TH850/TH1050 series is described below.

a) Axis 1
The output of axis 1 AC servo motor is input to the two-stage final drive helical gear. The output of the same gear directly controls the arm 1 motion.

b) Axis 2
The output of axis 2 AC servo motor is input to the two-stage final drive helical gear. The output of the same gear directly controls the arm 2 motion.

c) Axis 3
The output of axis 3 AC servo motor is input to the ball screw nut through the timing belt to move the tool shaft up and down.

d) Axis 4
The output of axis 4 AC servo motor is input to the ball spline nut through the two-stage reduction timing belt to revolve the tool shaft.

The layout of the robot mechanical components is shown in Fig. 2.2 which refers to the TH450/TH550.

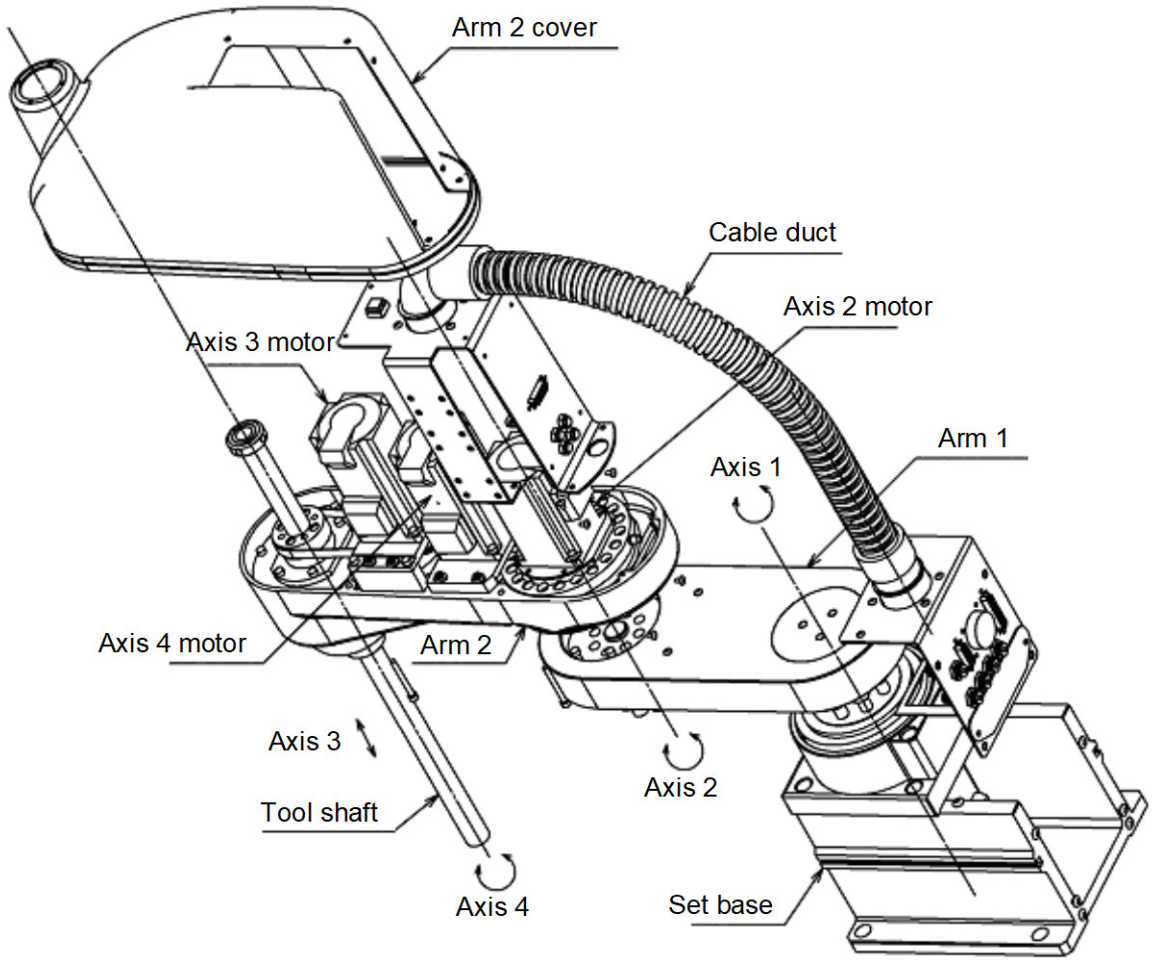


Fig. 2.2 Layout of robot mechanical components of TH450/TH550

The drive mechanism of each axis of the TH450/TH550 is described below.

a) Axis 1
The output of axis 1 AC servo motor is input to the harmonic drive. The output of the harmonic drive directly controls the arm 1 motion.

a) Axis 2
The output of axis 2 AC servo motor is input to the harmonic drive. The output of the harmonic drive directly controls the arm 2 motion.

c) Axis 3
The output of axis 3 AC servo motor is input to the ball screw nut through the timing belt to move the tool shaft up and down.

d) Axis 4
The output of axis 4 AC servo motor is input to the ball spline nut through the timing belt to revolve the tool shaft.

2.3 Robot Maintenance Procedures

2.3.1 Check of Cable Duct for Abnormality

The cable layout of the TH450/TH550 series is shown in Fig. 2.3.

a) Visually check to see that the cable duct is not locally twisted, bent or cracked when the robot moves. If there is any abnormality, replace the cable to prevent breakage of the cable wire, etc.

b) The connectors securing the cable duct are secured by tightening it to the plate with a hexagon nut. If the hexagon nut is loosened, tighten it with a pair of water pump pliers, etc.

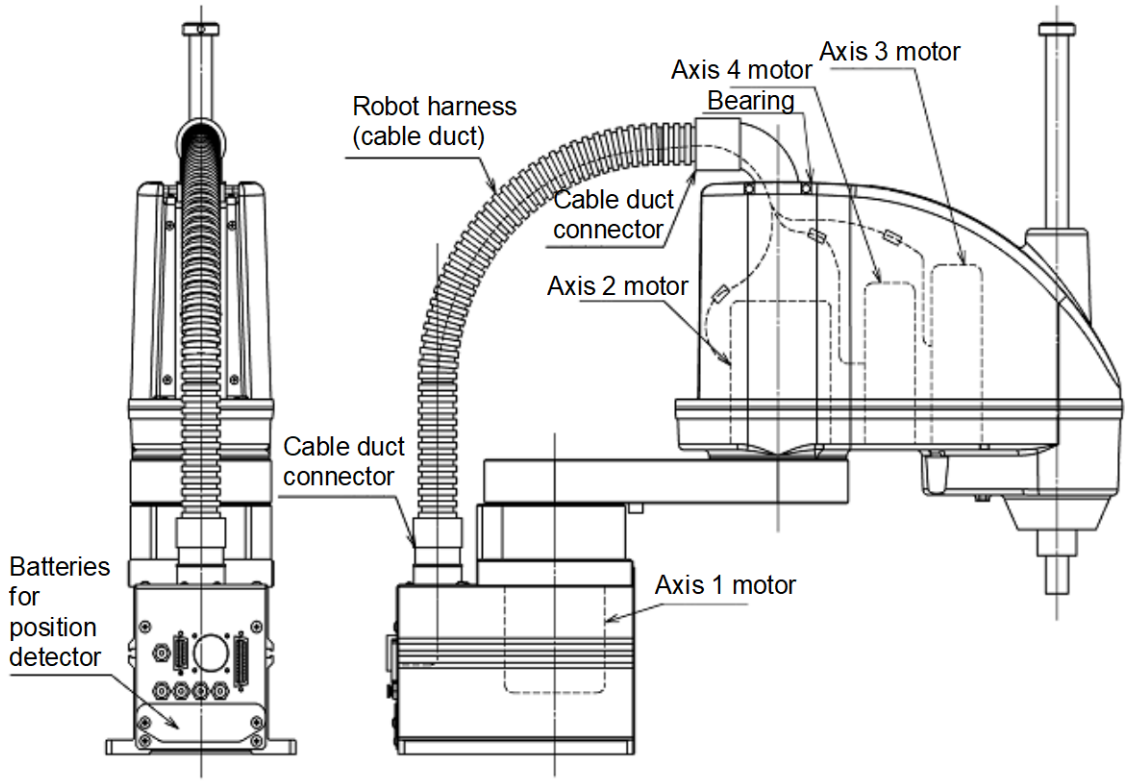


Fig. 2.3 TH450/TH550 series cable layout

2.3.2 Check of Cable for Abnormality

The cable layout of the TH650/TH850/TH1050 is shown in Fig. 2.4.

a) The cables pass from the robot base to the arm 1 and arm 2. As they pass through the robot interior, remove the arm 2 cover and arm 1 cover to check for an abnormality. Visually check to see that each cable is not locally twisted, bent or cracked when the robot moves.

b) The cable of the robot joint is most likely to break due to friction or twist. Visually check to see that the cable outer surface of the arm 1 and arm 2 is not damaged. If there is any abnormality, stop the robot operation and contact us immediately.

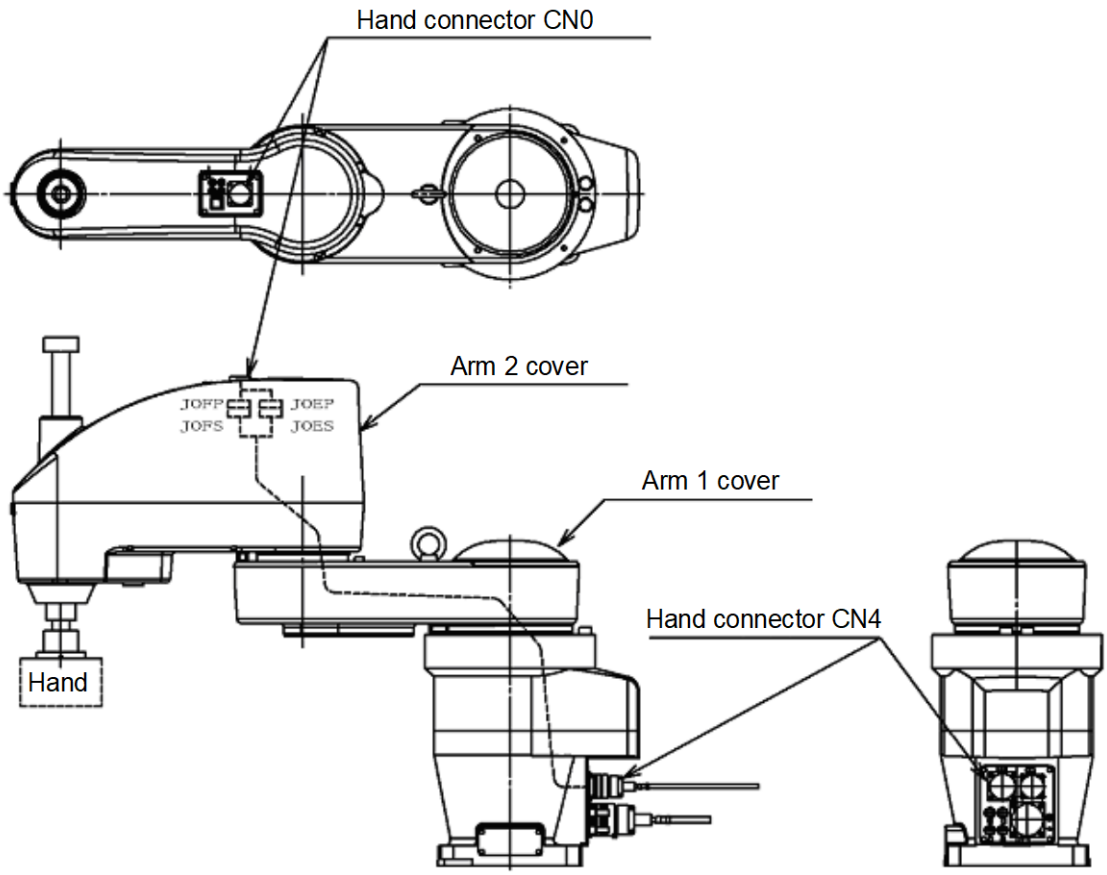


Fig. 2.4 TH650/TH850/TH1050 series cable layout

2.3.3 Check of Tool and Base for Mounting

DANGER

- Completely tighten the tool clamp bolt. Otherwise, the tool may drop or shift from a predetermined position.

a) Check of tool set flange bolt for looseness
Make sure, using a hexagonal wrench key, that the hexagon socket head cap screw of the tool set flange that is tightened onto the tool shaft is not loosened. Otherwise, tighten it.
Also make sure by means of a hexagonal wrench key that the hexagon socket head cap screw securing the tool set flange and tool is not loosened. Otherwise, tighten it.

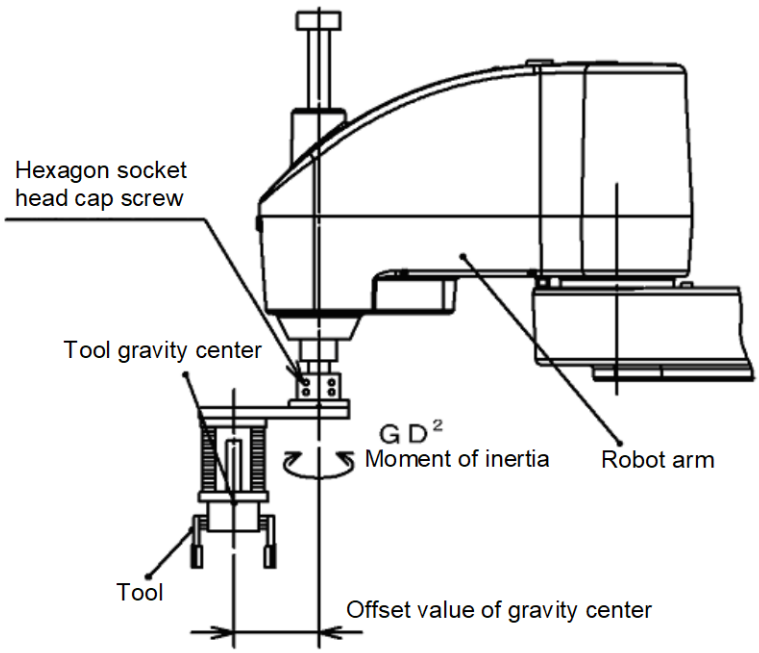


Fig. 2.5 Tool set area

b) Check of basic robot installation bolts
Make sure, using a hexagonal wrench key, that the installation bolts of the robot base are not loosened. Otherwise, tighten them.

2.3.4Check of Air Leakage

Piping connections in the robot are shown in Figs. 2.6 and 2.7. Make sure that no air leaks from the robot interior or air joints.

DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

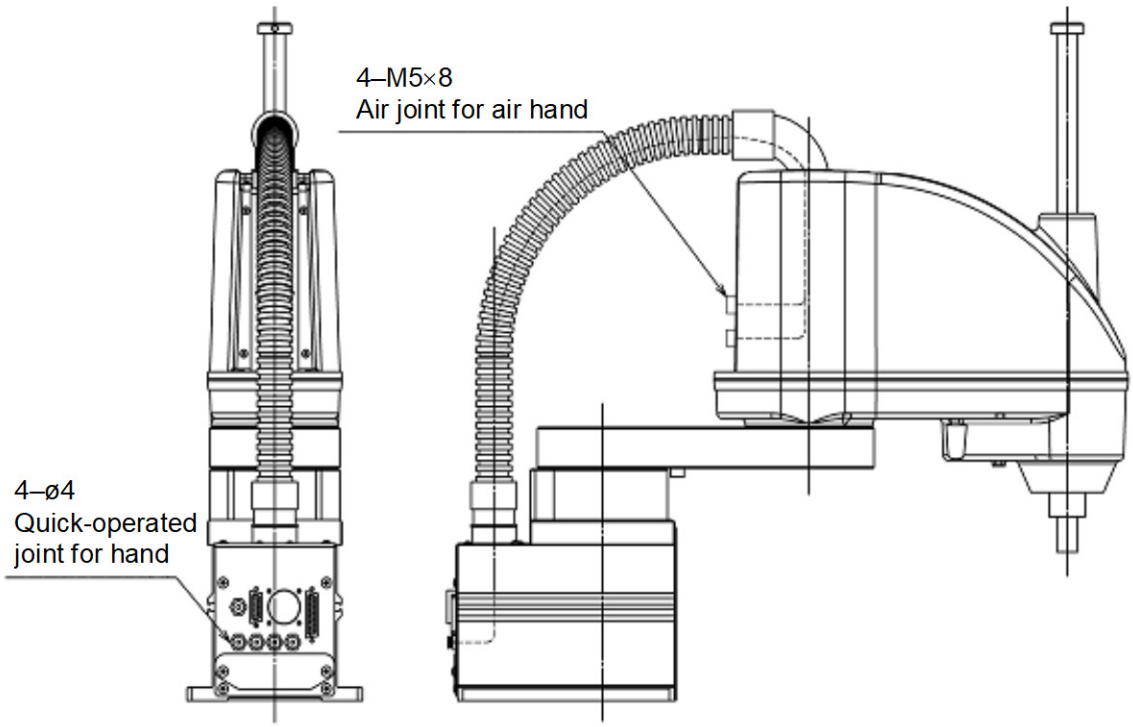


Fig. 2.6 TH450/TH550 tool air piping

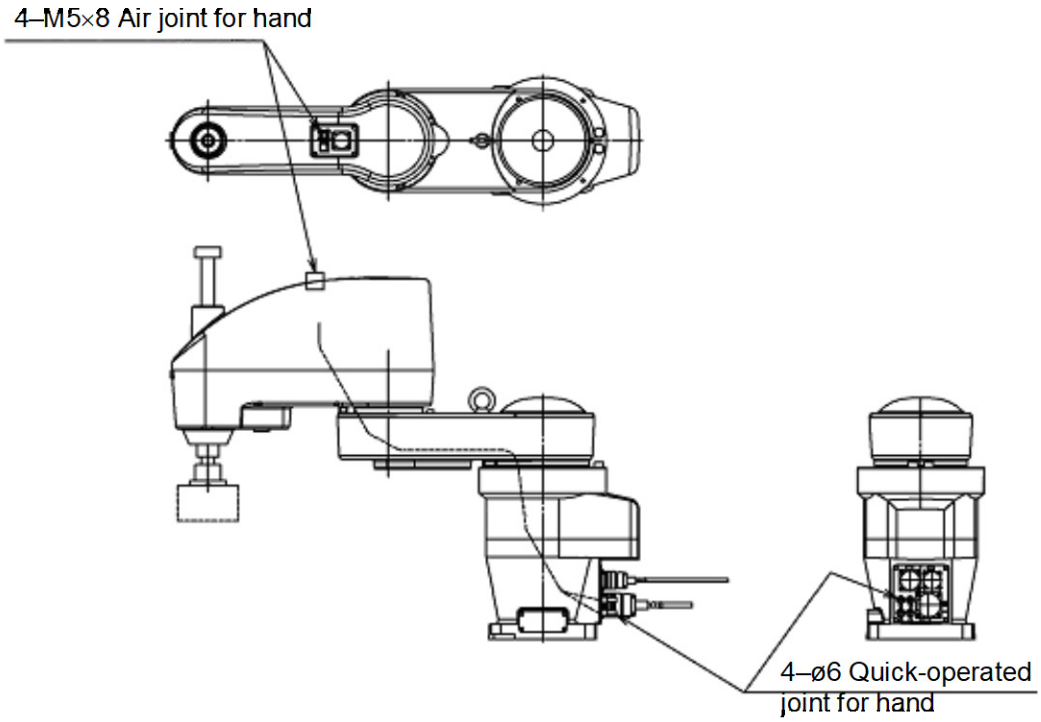


Fig. 2.7 TH650/TH850/TH1050 tool air piping

2.3.5 Lubrication of Tool Shaft

To lubricate the tool shaft, use the following grease.

- Lithium soap based grease No.2 (Ex. Alvania S2 made by Shell Oil)

! CAUTION

- DO NOT mistake the type of grease. Otherwise, the robot performance may deteriorate or the robot may malfunction.
- After you have touched the tool shaft by hand, be sure to apply the grease to it. Otherwise, rust will develop on the shaft.

a) Check of tool shaft for lubrication
Open the arm 2 cover and make sure that the tool shaft has not run out of grease and that no rust has developed on it. If the tool shaft has run out of grease, apply the grease according to Para. b).

b) Greasing of tool shaft
Apply the grease with a brush or by hand so that all guide slots around the shaft are covered. Just after the greasing, DO NOT start an operation. First, move the tool shaft slowly over the entire motion range. Then wipe off any grease that oozes from the shaft.

c) Greasing of tool shaft support bearings
Fill the specified volume of grease from the grease nipples at the end of the arm 2, using a grease gun. The greasing volume is 5 cm^3, and the greasing locations are shown in Fig. 2.8.

Type of grease nipple: A–M6F

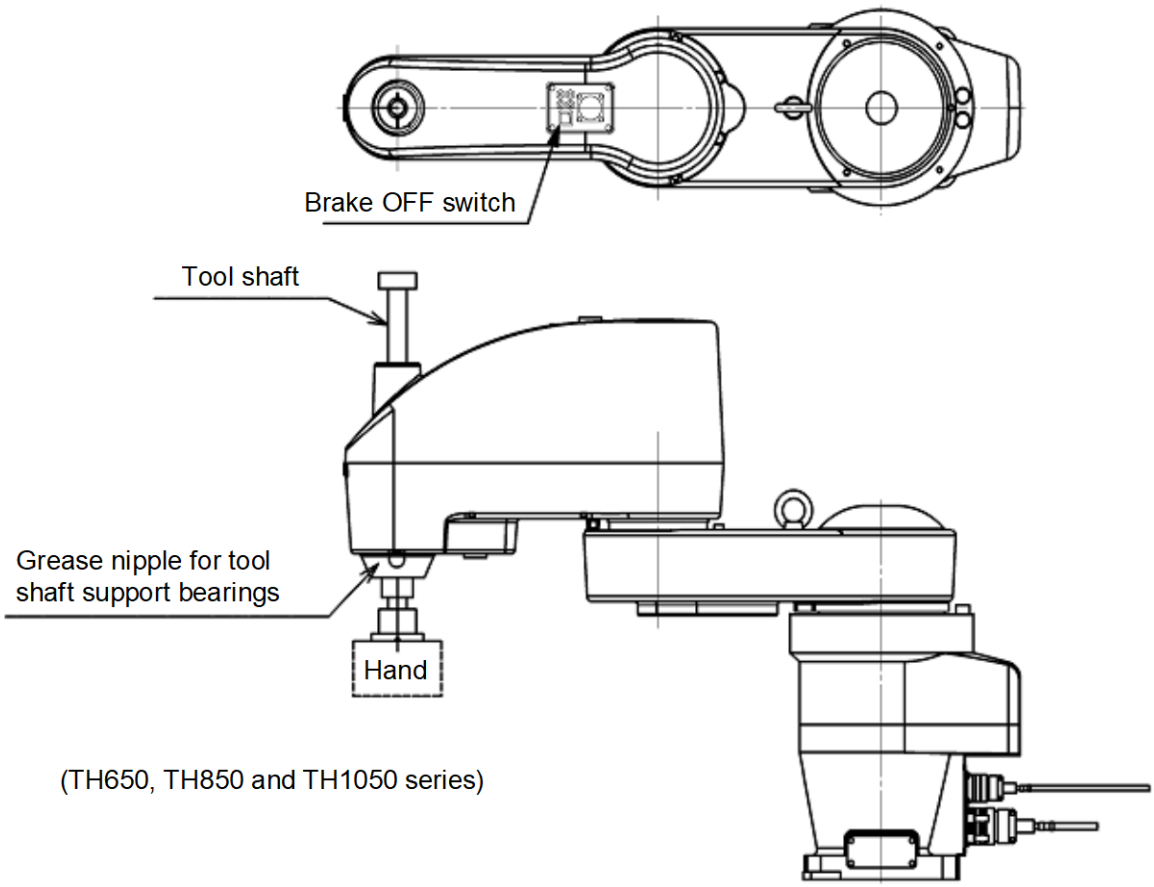


Fig. 2.8 Greasing locations of tool shaft support bearings

2.3.6 Check of Each Axis Drive Status

After turning on the main power switch of the controller, keep the EMERGENCY stop pushbutton switch in the depressed state.
Move each axis by hand to make sure that it can move smoothly.
When the brake OFF switch is pressed, the axis 3 and axis 4 enter a brake OFF state. Take careful precautions because the tool shaft may drop due to the weight of the hand and tool.

2.3.7 Check of Timing Belt Status

Remove the arm 2 cover, and visually check to see that the belts of axis 3 and axis 4 are not worn off or cracked.
Also check the timing belt for tension and adjust as necessary.

2.3.8 Greasing of Reduction Gears

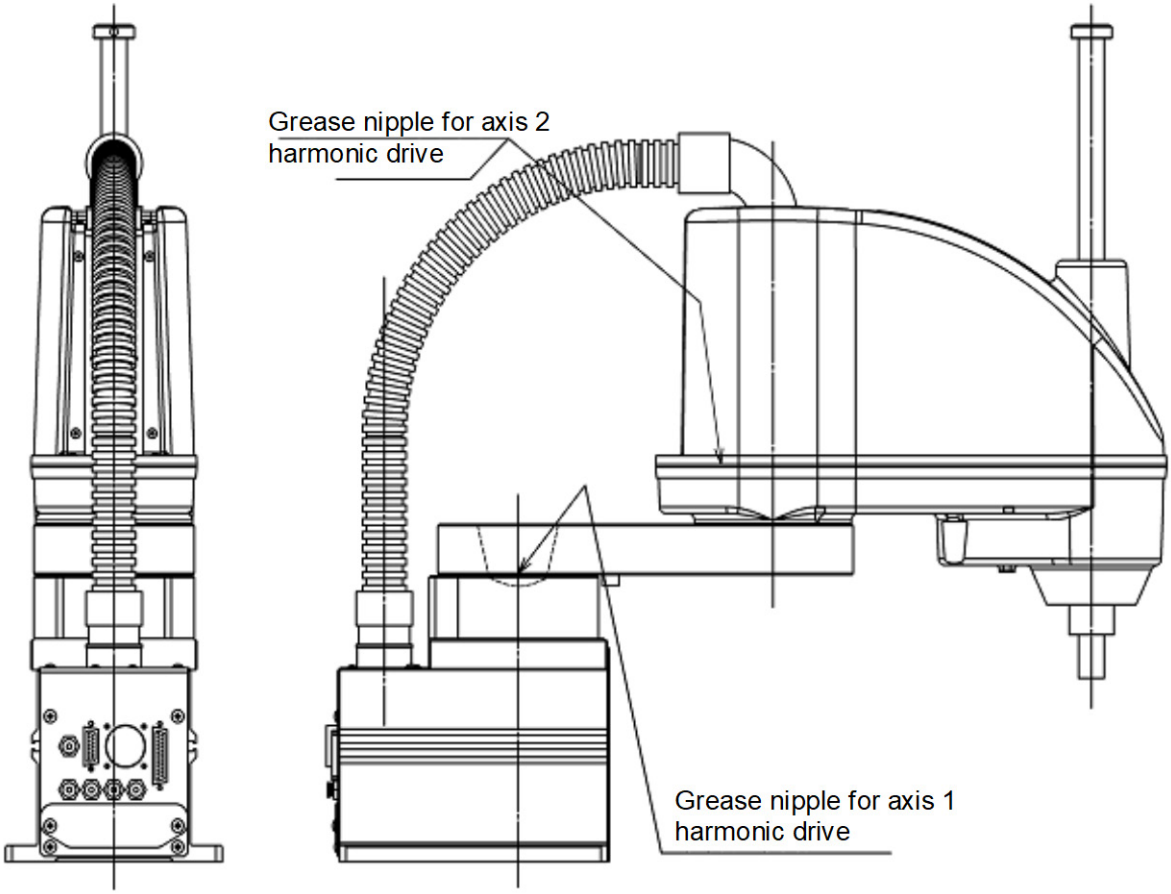
To lubricate the reduction gears, use the following grease.

- Harmonic grease SK-1A (Maker: Harmonic Drive Systems)

! CAUTION

- DO NOT mistake the type of grease. Otherwise, the robot performance may deteriorate or the robot may malfunction.

Remove the axis 1 cover and axis 2 end cover, and fill the specified volume of grease from the grease nipples, using a grease gun.
The greasing volume is 10 cm^3 (initial time 20 cm^3) and the greasing locations are shown in Fig. 2.9.
The type of grease nipple is A–M6F.



Type	Greasing location	Type of grease	Greasing volume
TH450	Axis 1 and axis 2	Harmonic grease SK-1A	10 cm^3 (Initial time 20 cm^3)
TH550	Axis 1 and axis 2		

Fig. 2.9 Greasing locations of reduction gears

2.3.9 Replacement of Position Detector (or Encoder) Batteries

To back up the data of the position detector attached to each motor, batteries (alkali dry batteries) are used. Replace the batteries every year.

Also if you do not use the robot for a long time (you turn off power of the robot), please replace the batteries when you turn on the robot. The battery replacement period is taken two months from the date that you turned off the power.

If the controller has detected the battery voltage drop alarm or the battery life nears the end (approximately two (2) years from the date of manufacture), replace the batteries at an early stage
The batteries should be replaced while the power is turned on and emergency stop is effected if the safety can be assured.

[Replacing procedures]

1) If the safety can be assured, press the EMERGENCY stop switch to emergency-stop the robot while the power is turned on. Unless the safety can be assured, turn off the main power switch.
2) Remove the cable connector set panel from the robot base unit.
3) Remove the battery box from the robot base unit.
4) Remove the battery box cover, then the batteries.
5) Set three (3) new batteries within five (5) minutes, taking care of the "+" and "–" polarities. When replacing the batteries while the power is turned off, wait for more than one (1) minute before replacement because the power failure working current flows for several tens seconds, then replace the batteries within five (5) minutes.
6) Set the battery box cover, and return the battery box in the robot base unit.
7) Mount the cable connector set panel again.

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

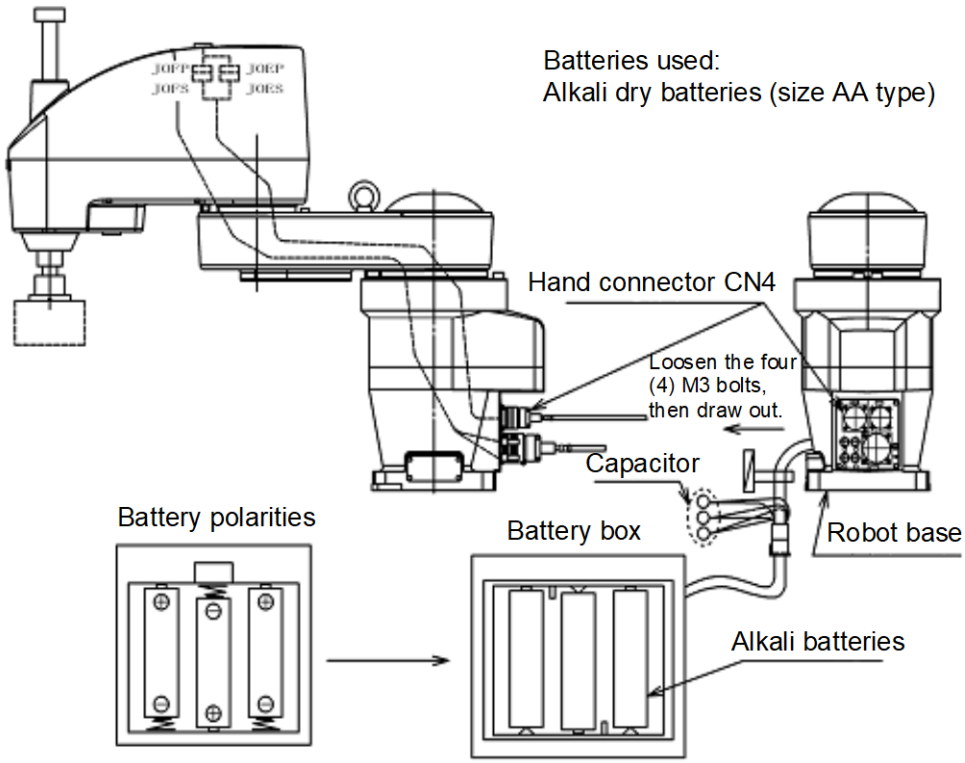


Fig. 2.10 Position detector battery set position

Section 3 Maintenance of Controller

3.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of robot controller, follow the items given below so that the work can be carried out safely.

! CAUTION

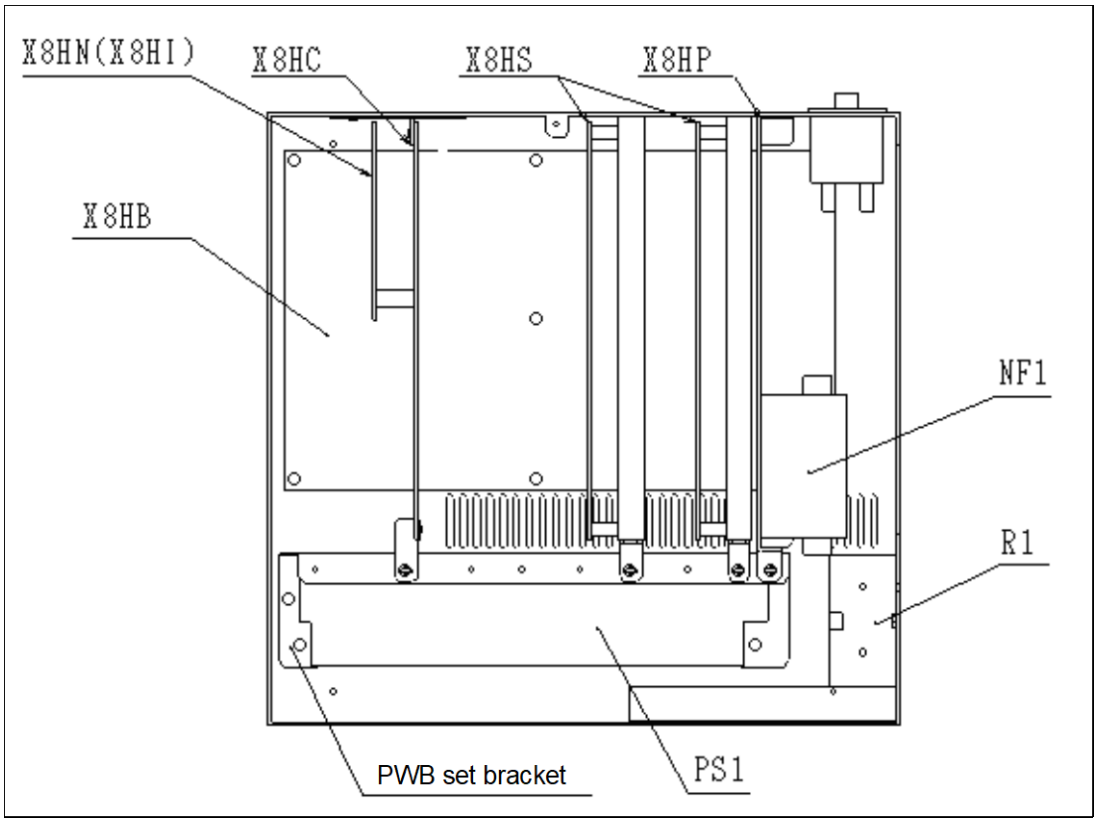
- Before removing the controller cover for maintenance or inspection, be sure to turn off the main power switch of the controller.
  Also, turn off the power breaker when connecting or disconnecting the power cable.
- DO NOT touch the servo driver for at least two (2) minutes after the main power switch is turned off. The large-capacity capacitor in the servo driver may be charged with voltage, and touching the servo driver could lead to electric shocks.
- DO NOT disconnect the battery connector at other than battery replacement. Otherwise, files saved in the memory may be lost.

! DANGER

Be sure to turn off the main power switch. If the main power switch remains ON, the servo power printed board, servo printed board and switching power supply are charged, and touching them could lead to electric shocks.

3.2 Layout of Controller Parts

3.2.1 Layout of TS2000 Parts

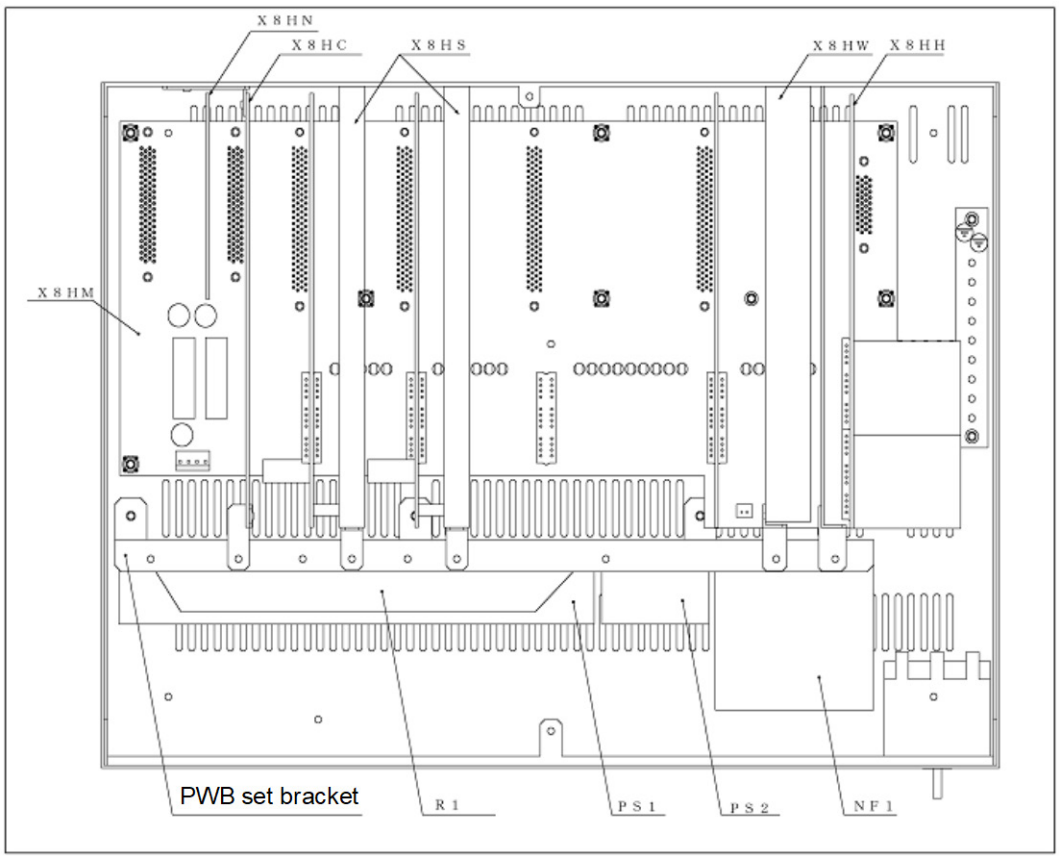


Upper side view

Fig. 3.1 Layout of TS2000 controller parts

Part name	Descriptions
PS1	P5V, P24V switching power supply
PWB set bracket	PS1 set bracket
X8HP	Servo power printed board
X8HC	Main control printed board
X8HN (X8HI)	I/O output printed board
X8HS	Servo printed board
X8HB	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.2.2 Layout of TS2100 Parts



Upper side view

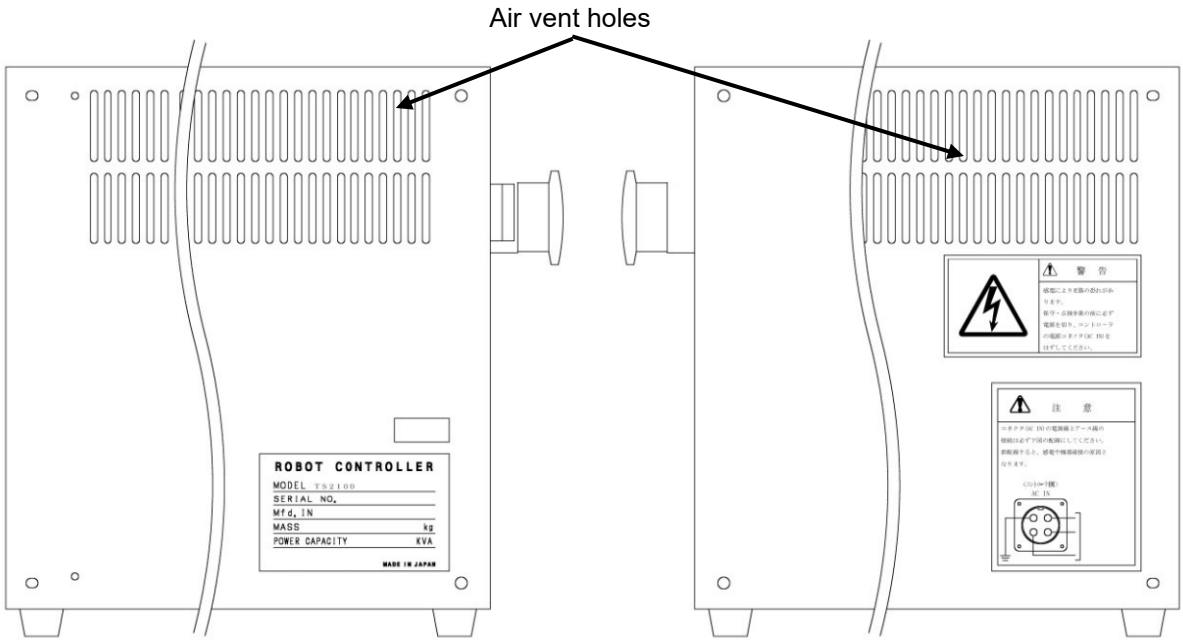
Fig. 3.2 Layout of TS2100 controller parts

Part name	Descriptions
PS1	P5V, P24V switching power supply
PS2	P24V switching power supply
PWB set bracket	PS1 set bracket
X8HH	Servo power printed board
X8HC	Main control printed board
X8HN (X8HI)	I/O output printed board
X8HS, X8HW	Servo printed board
X8HM	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.3 Maintenance Procedures

3.3.1 Check of Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction. To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.



Left side view Right side view

Fig. 3.3 Controller side views

a) If there is any obstacles, move them away from the vent holes so that the air flow is not blocked.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is let on the vent hole, remove it.

c) If the controller is placed sideways, make sure that the air vent holes on the bottom surface of the controller are more than 11 mm away from the floor surface. Always keep the space of 11 mm or over. The length of rubber shoe is approximately 11 mm.

3.3.2 Check of Safety Devices for Function

Make sure that the emergency stop pushbutton switches equipped on the control panel and teach pendant work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

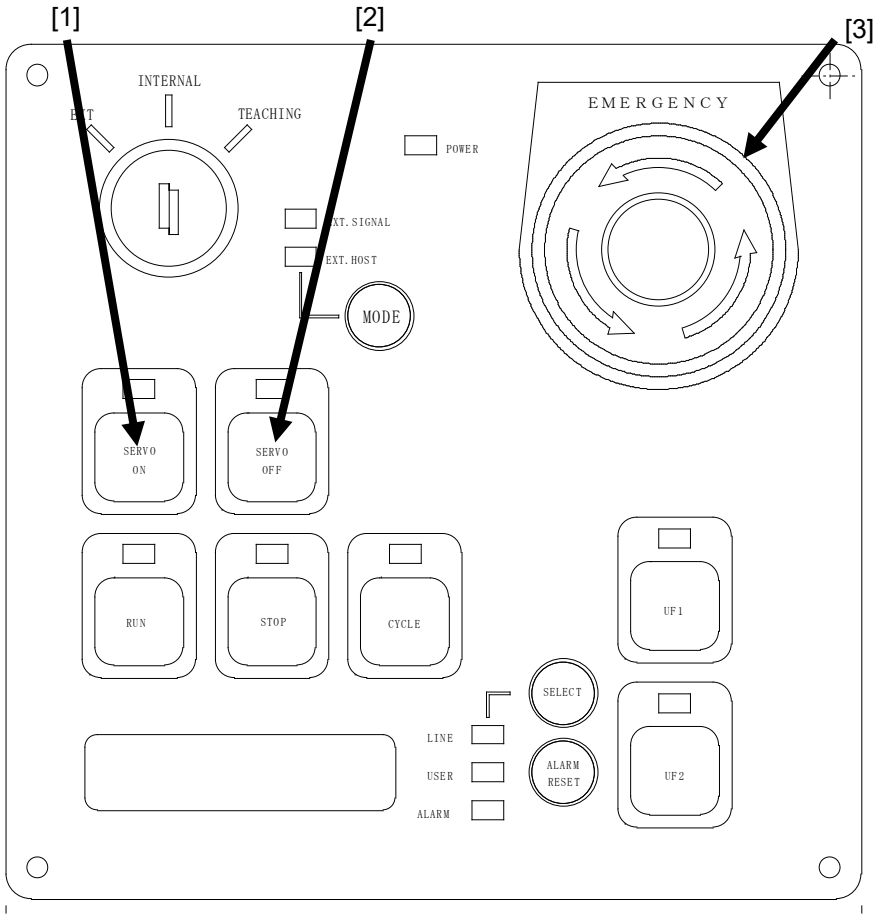


Fig. 3.4 Layout of control panel switches

a) Make sure that the EMERGENCY (emergency stop) switch [2] provided on the control panel functions properly.

1) Press the SERVO ON switch [1] provided on the control panel to turn the servos on and make sure that the SERVO ON switch [1] LED is illuminated.

2) Press the EMERGENCY switch [2] and make sure that the servos are turned off. The SERVO ON switch [1] LED goes out and the SERVO OFF switch [3] LED is illuminated. At this time, make sure that the EMERGENCY switch [2] remains depressed.

3) Turn the EMERGENCY switch [2] clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY STOP switch [6] provided on the teach pendant functions properly.

1) Press the SERVO ON switch [4] on the teach pendant to turn on the servo power, and make sure that the SERVO ON switch LED [5] is illuminated.

Note: The servo power can be turned on only in the TEST RUN mode.

2) Press the EMERGENCY STOP switch [6] on the teach pendant and make sure that the servos are turned off. The SERVO ON switch LED [5] goes out.

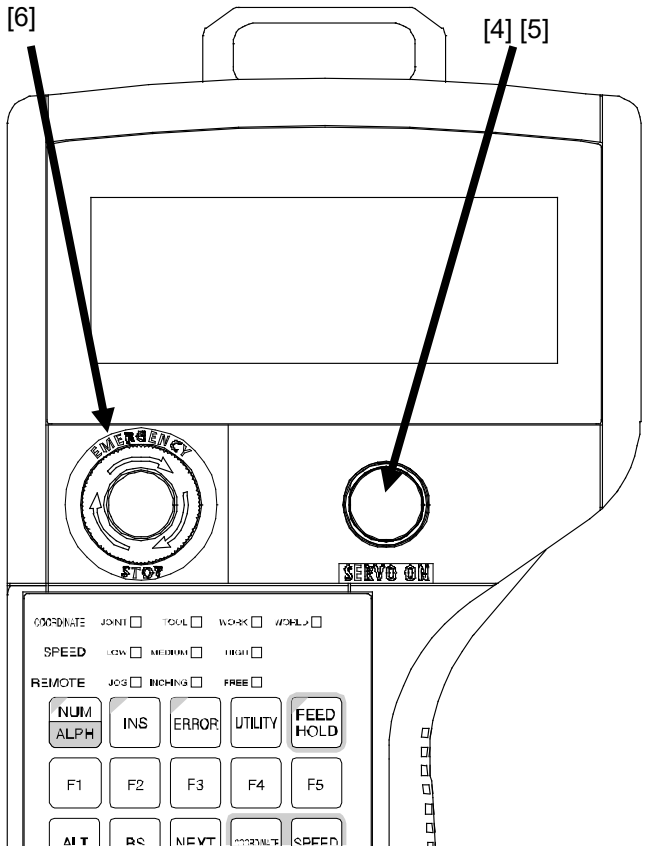


Fig 3.5 Teach pendant EMERGENCY STOP switch

At this time, make sure that the EMERGENCY STOP switch [6] remains depressed.

3) Turn the EMERGENCY STOP switch [6] clockwise and make sure that the same switch turns off.

c) Make sure that the safety devices controlled by the external operation signals work properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal "emergency stop" (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal "low speed command", and make sure that each signal functions properly.

3.3.3 Battery Replacement

SRAM equipped on the X8HC printed board of the TS2000 /TS2100 robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the SRAM data, resulting in data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

- 1–145 MAIN Battery alarm

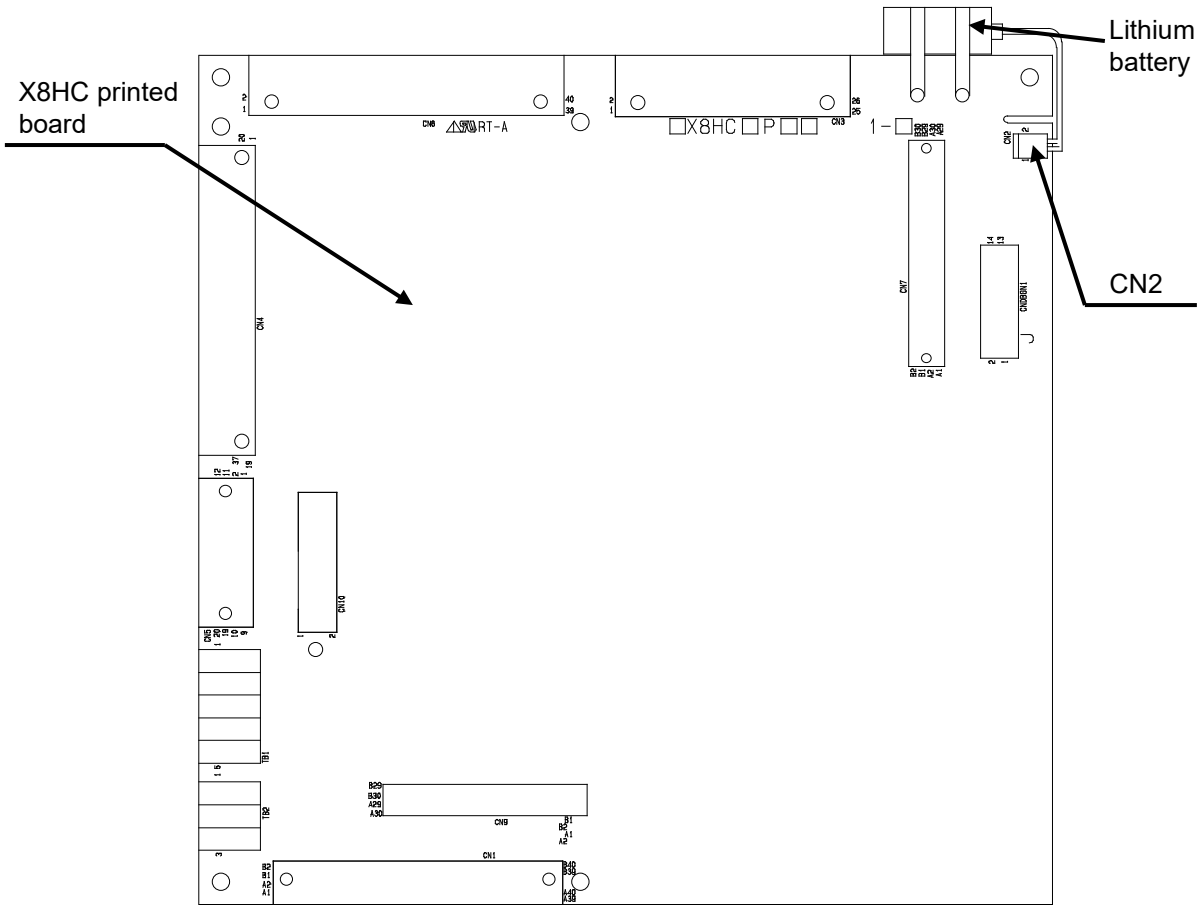


Fig. 3.6 Location of lithium battery

Table 3.1 Type of battery

Type of battery	Manufacturer	Battery service life
ER3VC	Toshiba Denchi	10 years (at room temperature)

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.

Only the battery shown in Table 3.1 should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) If the battery is to be kept removed for more than five (5) minutes, copy all programs stored in the SRAM to the personal computer. In the worst case, all memory data are lost.

2) Set OFF the main power switch.

3) Remove the cover from the body.

4) Remove the battery connector from CN2 of the X8HC printed board, then disconnect the battery from the Tie-wrap securing the battery.
(For the battery position, see Fig. 3.5.)

5) Set a new battery within five (5) minutes.
Even if not used, the lithium battery capacity decreases by natural discharge. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

6) Mount the cover on the body.

7) If the battery is kept removed for more than five (5) minutes, turn on the main power switch and load the programs from the floppy disks.

If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

! CAUTION

- Waste battery should be disposed of according to the user's in-house regulations.
- NEVER drop the battery into fire. NEVER short-circuit, charge, disassemble or heat it. Otherwise, liquid leakage or rupture may be caused.

3.3.4 Replacement of Switching Power Supply Unit

The life of the switching power supply unit (DC5V/DC24V) used in the TS2000/TS2100 controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large, the running time is long and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[Replacement procedures]

1) Turn off the main power switch.

2) Turn off the power breaker.

3) Remove the cover from the body.

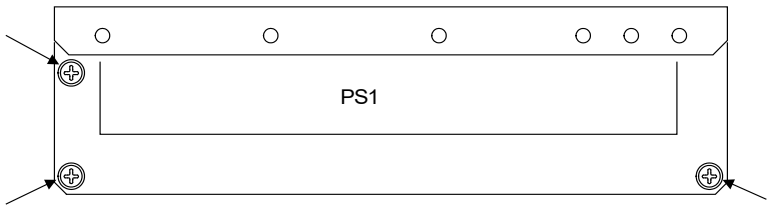
4) Remove all connectors in the control panel, which are linked to the X8HC, X8HS (2 pcs.), X8HP, X8HH and X8HW printed boards. (For the position of each printed board, see Fig. 3.1 and Fig. 3.2. The X8HH and X8HW printed boards are used for TS2100.)

5) Remove the X8HC, X8HN (X8HI), X8HS (2 pcs.), X8HP, X8HH and X8HW printed boards from the body. (The X8HH and X8HW printed boards are used for TS2100.)

6) Disconnect the cable connecting the parts on the PWB set bracket.

7) Remove the screws (3 places) which secure the body and PWB set bracket. (For the position of the PWB set bracket, see Fig. 3.1 and Fig. 3.2.)

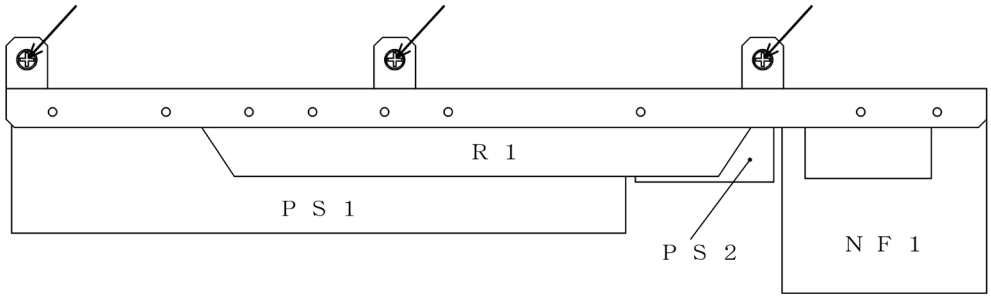
- TS2000 PWB set bracket



Upper side view

Fig. 3.7 TS2000 PWB set bracket (upper side view)

- TS2100 PWB set bracket



Upper side view

Fig. 3.8 TS2100 PWB set bracket (upper side view)

8) Remove the PWB set bracket from the body.

9) Remove each power supply from the PWB set bracket, and set a new power supply.

- TS2000 PWB set bracket

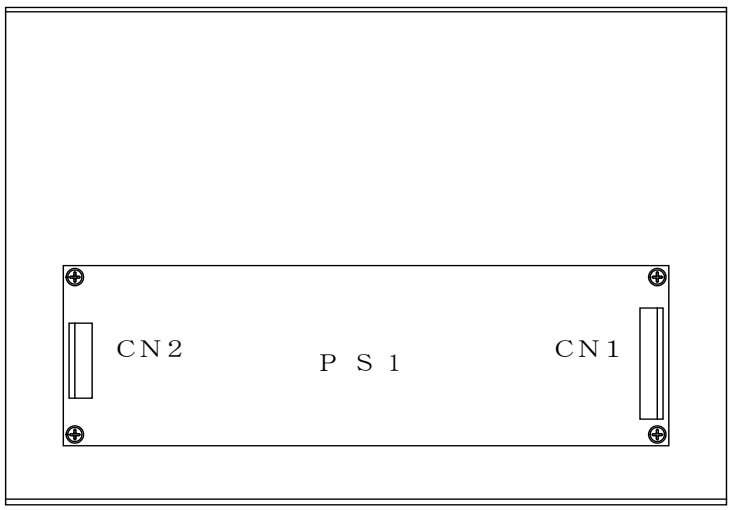


Fig. 3.9 TS2000 PWB set bracket

- TS2100 PWB set bracket

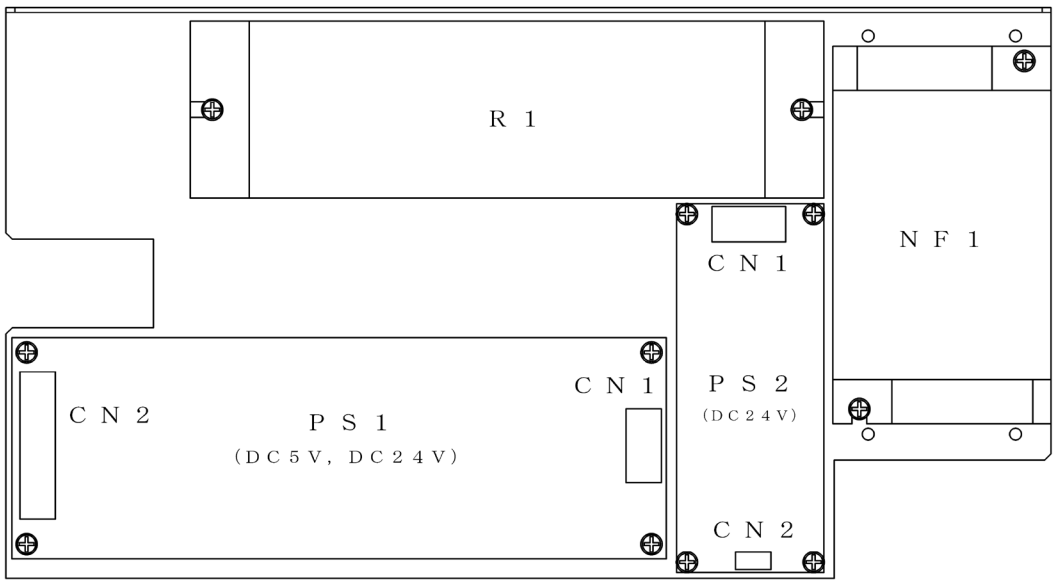


Fig. 3.10 TS2100 PWB set bracket

10) Restore the PWB set bracket to the body, where the new power supply is set. Then link the connector.

11) Screw the PWB set bracket into the body (three (3) places; See Fig. 3.7 and Fig. 3.8.)

12) Mount the removed printed boards on the body.

13) Mount the removed connectors on respective printed boards.

14) Mount the cover on the body.

3.3.5 Replacement of Fuse (X8HN, H8HI Printed Boards)

If the current exceeding the specified current has run through the I/O unit, the fuse of the X8HN (X8HI) printed board is blown out. If the alarm saying "I/O FUSE Broken (8–273)" is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.

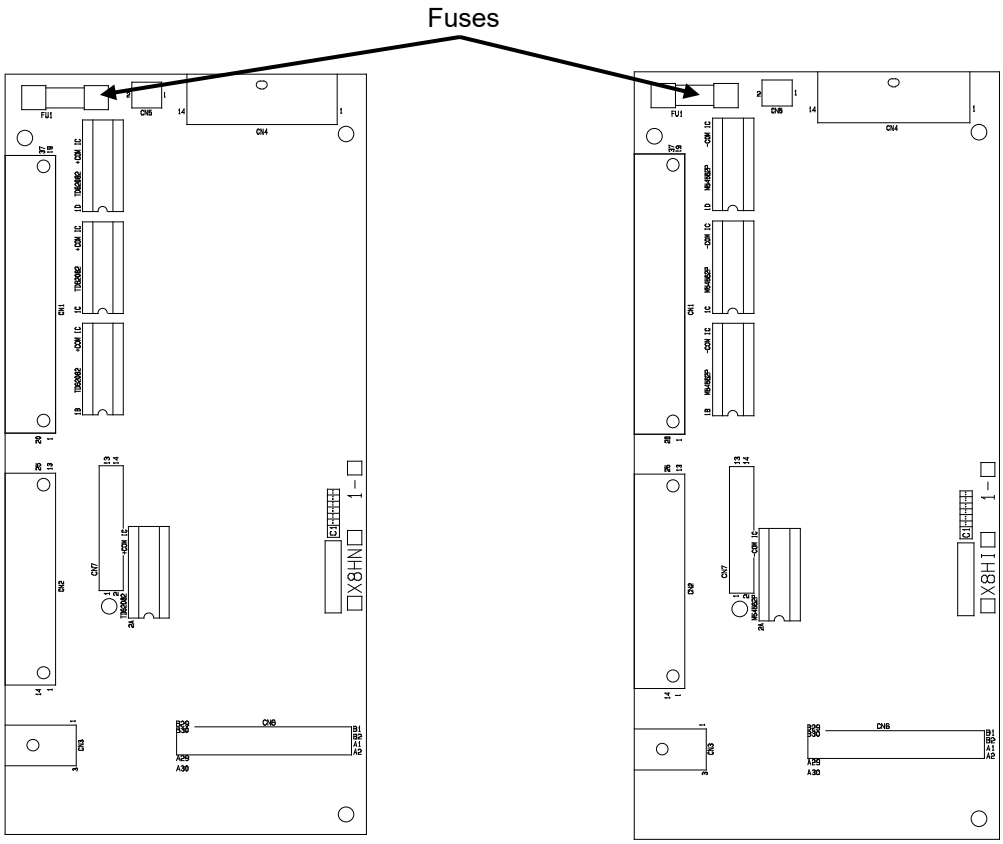


Table 3.1a Type of fuse

Type of fuse	Manufacturer
51NM030H	PICO

[Replacement procedures]

1) Turn off the controller main power.

2) Remove the cover attached to the fuse unit as shown above.

3) Remove the fuse and set a new fuse.

4) Mount the removed cover.

5) Turn on the controller main power and make sure that no error will occur.

3.3.6 Replacement of Output ICs (X8HN, H8HI Printed Boards)

If the current exceeding the specified current has run through the output unit, the outputs may not turn on normally.
When this happens, the ICs of the X8HN (X8HI) printed board have possibly been damaged. Replace the ICs.
Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.

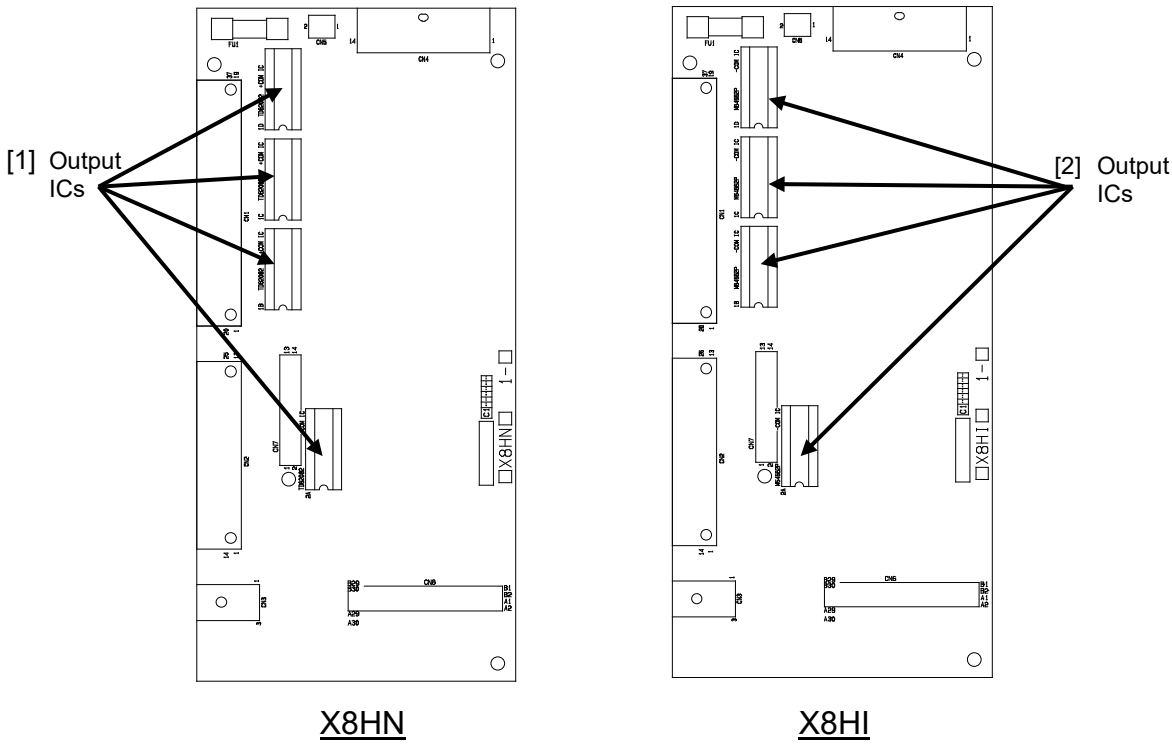


Table 3.1b Type of IC

	Name of board	Type of IC	Manufacturer
[1]	X8HN	TD62082AP	Toshiba Corp.
[2]	X8HI	M54562P	Mitsubishi Electric

[Replacement procedures]

1) Turn off the controller main power.

2) Remove the cover from the body.

3) Remove all connectors in the control panel, which are linked to the X8HC printed board.

4) Remove the X8HC and X8HN (X8HI) printed boards from the body.

5) Remove the ICs shown above from respective sockets.

6) Attach new ICs to the sockets.

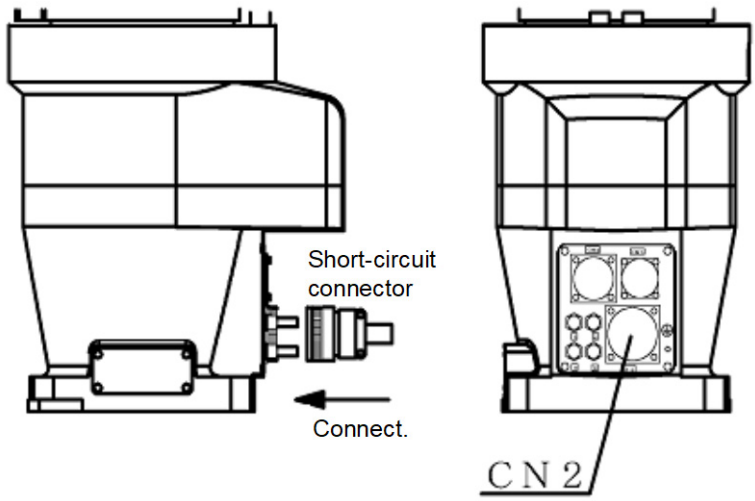
7) Mount the X8HC and X8HN (X8HI) printed boards on the body.

8) Link the removed connectors to the X8HC printed board, and set the main cover.

9) Turn on the controller main power.

3.3.7 Cautions on Transport

When the robot power cable has been disconnected from the controller, attach the following short-circuit connector (accessory) to the robot power cable.
If the robot has been moved suddenly or heavy vibration has been imposed on the robot at power OFF, without connecting the short-circuit connector, the absolute coordinate counter of the encoder cannot trace any further, resulting in generation of a position sensor error or loss of origin position data.



Attachment 1. List of Mechanical Replacement Parts for Maintenance

The list of robot replacement parts for maintenance is shown in the attachment 1 and attachment 1.1~1.3, and those locations are shown in attachment figure A.

List of TH450/TH550 replacement parts for maintenance

Part name	Location	Type [TH550]	Unit Code	Maker	Q'ty	Remarks
AC servo motor	A1	MSMD092S2U	Y610A3400	Panasonic Motor Co.	1	Axis 1
	A2	MSMD042S2U	Y610A34C0		1	Axis 2
	A3	MSMD022S2V	Y610A3430		2	Axis3, axis4
Harmonic drive	B1	M147090 [M153235]	Y610A34L0 [Y610A34N0]	SHIBAURA MACHINE CO.,LTD.	1	Axis 1
	B2	M147091	Y610A34M0		1	Axis 2
Timing belt	C1	345-3GT-9	-		1	Axis 3
	C2	333-3GT-9	-		1	Axis 4
	C3	360-5GT-20	-		1	
Timing pulley	D1	M147308	-		1	Axis 3
	D2	M147309	-		1	Axis 3
	D3	M147310	-		1	Axis 4
	D4	M147312	-		1	
	D5	M147313	-		1	Axis 4
	D6	M147314	-		1	
Ball screw spline	E1	M147099	Y610A34G0		1	Axis 3
		M153034	Y610A34H0		1	Axis 3(1)
Harness	F1	F090886 [F091003]	Y610A33H0 [Y610A33S0]		1	-
Connector for short-circuit	-	M152934	Y610A34S0		1	Option
Grease	-	Harmonic grease SK-1A	-	Harmonic Drive Systems	500g 2.5kg	-
	-	Shell Alvania grease S2	-	Shell Oil	2.5kg	-
Position detector battery	G1	Alkali dry batteries (Size AA type)	-	-	3	Normal
	G1	Lithium battery	Y610A33T0	SHIBAURA MACHINE CO.,LTD.	1	Long life type Option

(1) : Option of the stroke 300mm

・Please contact SHIBAURA MACHINE at order entry, after you confirm the serial number of the main robot.

! CAUTION

- With the exception of the Shell Alvania grease No.2, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts by the user.

List of TH450-T/TH550-T (Ceiling Type) replacement parts for maintenance

Part name	Location	Type [TH550-T]	Unit Code [TH550-T]	Maker	Q'ty	Remarks
AC servo motor	A1	MSMD092S2U	Y610A3400	Panasonic Motor Co.	1	Axis 1
	A2	MSMD042S2U	Y610A34C0		1	Axis 2
	A3	MSMD022S2V	Y610A3430		2	Axis 3, axis 4
Harmonic drive	B1	M147090 [M153235]	Y610A34L0 [Y610A34N0]	SHIBAURA MACHINE CO.,LTD.	1	Axis 1
	B2	M147091	Y610A34M0		1	Axis 2
Timing belt	C1	345-3GT-9	-		1	Axis 3
	C2	333-3GT-9	-		1	Axis 4
	C3	360-5GT-20	-		1	
Timing pulley	D1	M147308	-		1	Axis 3
	D2	M147309	-		1	Axis 3
	D3	M147310	-		1	Axis 4
	D4	M147312	-		1	
	D5	M147313	-		1	Axis 4
	D6	M147314	-		1	
Ball screw spline	E1	M147099	Y610A34G0		1	Axis 3
		M153034	Y610A34H0		1	Axis 3(1)
Harness	F1	F097797 [F097853]	Y610A35P0 [Y610A35V0]		1	-
Connector for short-circuit	-	M152934	Y610A34S0		1	Option
Grease	-	Harmonic grease SK-1A	-	Harmonic Drive Systems	500g 2.5kg	-
	-	Shell Alvania grease S2	-	Shell Oil	2.5kg	-
Position detector battery	G1	Alkali dry batteries (Size AA type)	-	-	3	Normal
	G1	Lithium battery	Y610A33T0	SHIBAURA MACHINE CO.,LTD.	1	Long life type Option

(1) : Option of the stroke 300mm

・Please contact SHIBAURA MACHINE at order entry, after you confirm the serial number of the main robot.

! CAUTION

- With the exception of the Shell Alvania grease No.2, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts by the user.

List of TH650 replacement parts for maintenance

Part name	Location	Type	Unit Code	Maker	Q'ty	Remarks
AC servo motor	A1	MSMD092S2U	Y610A3400	Panasonic Motor Co.	1	Axis 1
	A2	MSMD062S2U	Y610A3410		1	Axis 2
	A3	MSMD042S2V	Y610A3420		2	Axis 3, axis 4
Helical gear	B1	M198669	Y610A32X0	SHIBAURA MACHINE CO.,LTD.	1	Axis 1, number of teeth 9
	B2	M198670			1	Axis 1, number of teeth 95
	B3	M198671			1	Axis 1, number of teeth 12
	B4	H830644			1	Axis 1, number of teeth 91
	B5	M198672			1	Axis 2, number of teeth 8
	B6	M198674			1	Axis 2, number of teeth 74
	B7	M198673			1	Axis 2, number of teeth 13
	B8	H830645			1	Axis 2, number of teeth 71
Timing belt	Attachment 1.1				Attachment 1.1	
Timing pulley						
Ball screw spline	E1	H812079	Y610A3380		1	Axis 3
		H811850	Y610A33Q0		1	Axis 3(2)
Harness	-	F14202G01	Y610A3340		1	
Grease	-	Attachment 1.2				
	-	Shell Alvania grease S2	-	Shell Oil	2.5kg	-
Position detector battery	G1	Alkali dry batteries (Size AA type)	-	-	3	Normal
	G1	Lithium battery	Y610A33U0	SHIBAURA MACHINE CO.,LTD.	1	Long life type Option

(2) : Option of the stroke 400mm

・Please contact SHIBAURA MACHINE at order entry, after you confirm the serial number of the main robot.

Attachment 1.1 Timing belt and timing pulley of the TH650

Part name	Location	Type	Unit Code	Q'ty	Remarks
Timing belt	C1	432-3GT-12	-	1	Axis 3
	C2	378-3GT-20		1	Axis 4
	C3	450-5GT-40		1	
Timing pulley	D1	M126269		1	Axis 3
	D2	M126266		1	
	D3	M147189		1	Axis 4
	D4	M126265		1	
	D5	M118313		1	
	D6	M118310		1	
Timing belt	C1	489-3GT-12	Y610A32Y0	1	Axis 3
	C2	378-3GT-20		1	Axis 4
	C3	430-5GT-40		1	
Timing pulley	D1	M126269		1	Axis 3
	D2	M198379		1	
	D3	M147189		1	Axis 4
	D4	M126265		1	
	D5	M118313		1	
	D6	M198380		1	

・Please contact SHIBAURA MACHINE at order entry, after you confirm the serial number of the main robot.

Attachment 1.2 Grease of TH650/TH850/TH1050

Type	Maker	Q'ty	Remarks
G501	Shin-Etsu Chemical Co., Ltd.	400g	White
Dinahmax NO.2	Cosmo Oil Lubricants Co., Ltd.	500g/2.5kg	Beige

・Please contact SHIBAURA MACHINE at order entry, after you confirm the serial number of the main robot.

! CAUTION

- With the exception of the grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts by the user.
- Please do not make a mistake in the kind of the grease.

List of TH850/TH1050 replacement parts for maintenance

Part name	Location	Type [TH1050]	Unit Code	Maker	Q'ty	Remarks
AC servo motor	A1、A2	MSMD092S2U	Y610A3400	Panasonic Motor Co.	2	Axis 1, axis 2
	A3	MSMD062S2V	Y610A34B0		2	Axis 3, axis 4
Helical gear	B1	M198669	Y610A32Z0	SHIBAURA MACHINE CO.,LTD.	1	Axis 1, number of teeth 9
	B2	M208813			1	Axis 1, number of teeth 95
	B3	M208812			1	Axis 1, number of teeth 10
	B4	H830644			1	Axis 1, number of teeth 91
	B5	M198672			1	Axis 2, number of teeth 8
	B6	M208814			1	Axis 2, number of teeth 74
	B7	M208815			1	Axis 2, number of teeth 10
	B8	H830645			1	Axis 2, number of teeth 71
Timing belt	Attachment 1.3				Attachment 1.3	
Timing pulley						
Ball screw spline	E1	H812079	Y610A3380		1	Axis 3
		H811850	Y610A33Q0		1	Axis 3(2)
Harness	-	F14205G01 [F14204G01]	Y610A3530 [Y610A3540]		1	-
Grease	-	Attachment 1.2				
	-	Shell Alvania grease S2		Shell Oil	2.5kg	-
Position detector battery	G1	Alkali dry batteries (Size AA type)	-	-	3	Normal
	G1	Lithium battery	Y610A33U0	SHIBAURA MACHINE CO.,LTD.	1	Long life type Option

(2) : Option of the stroke 400mm

・Please contact SHIBAURA MACHINE at order entry, after you confirm the serial number of the main robot.

Attachment 1.3 Timing belt and timing pulley of the TH850/TH1050

Part name	Location	Type	Unit Code	Q'ty	Remarks
Timing belt	C1	750-3GT-15	-	1	Axis 3
	C2	580-5GT-20		1	Axis 4
	C3	615-5GT-40		1	
Timing pulley	D1	M153184		1	Axis 3
	D2	M153183		1	
	D3	M153019		1	Axis 4
	D4	M160634		1	
	D5	M153181		1	
	D6	M153182		1	
Timing belt	C1	720-3GT-15	Y610A32W0	1	Axis 3
	C2	580-5GT-20		1	Axis 4
	C3	615-5GT-40		1	
Timing pulley	D1	M153181		1	Axis 3
	D2	M208866		1	
	D3	M153019		1	Axis 4
	D4	M160634		1	
	D5	M153181		1	
	D6	M208865		1	

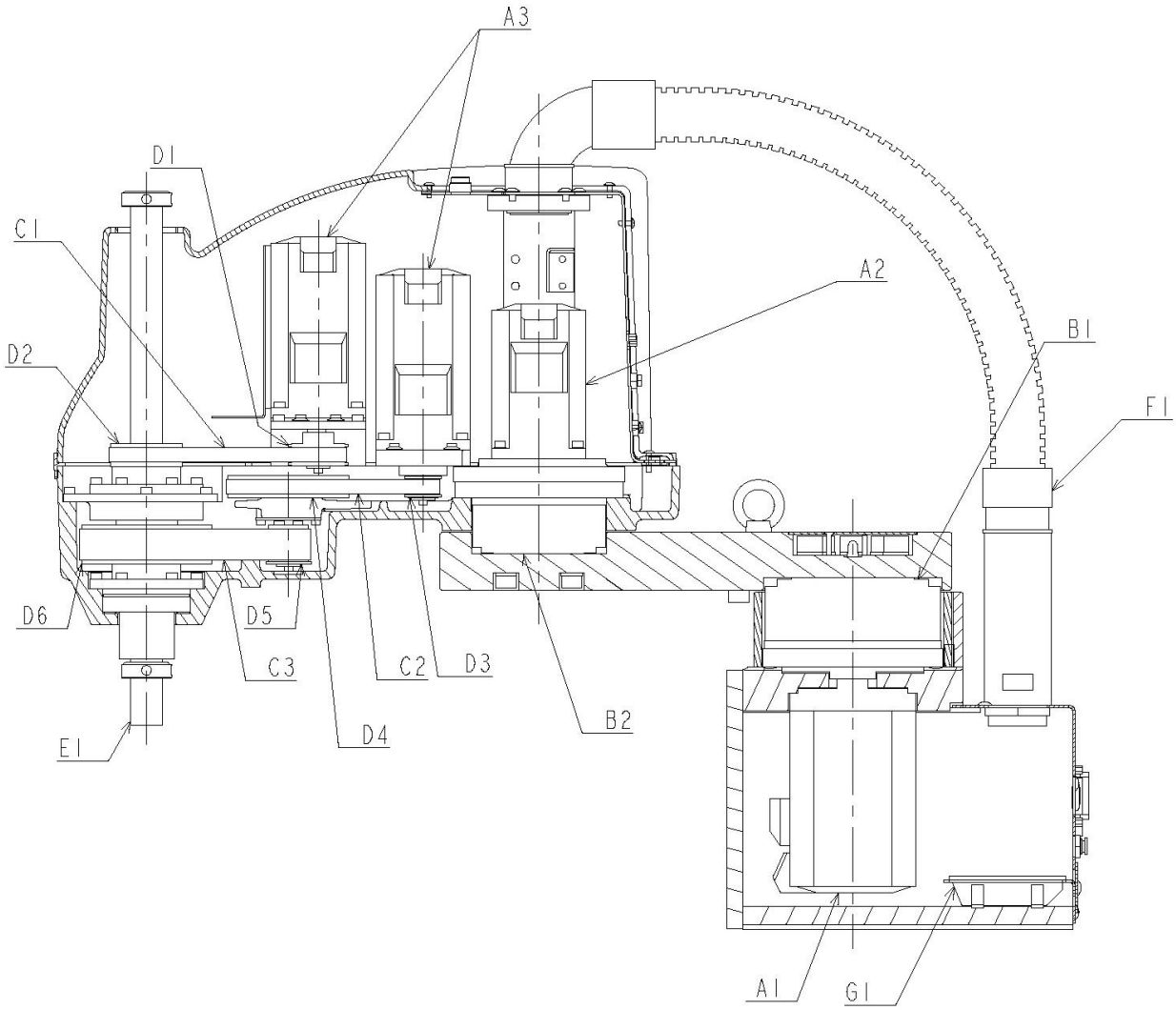
・Please contact SHIBAURA MACHINE at order entry, after you confirm the serial number of the main robot.

! CAUTION

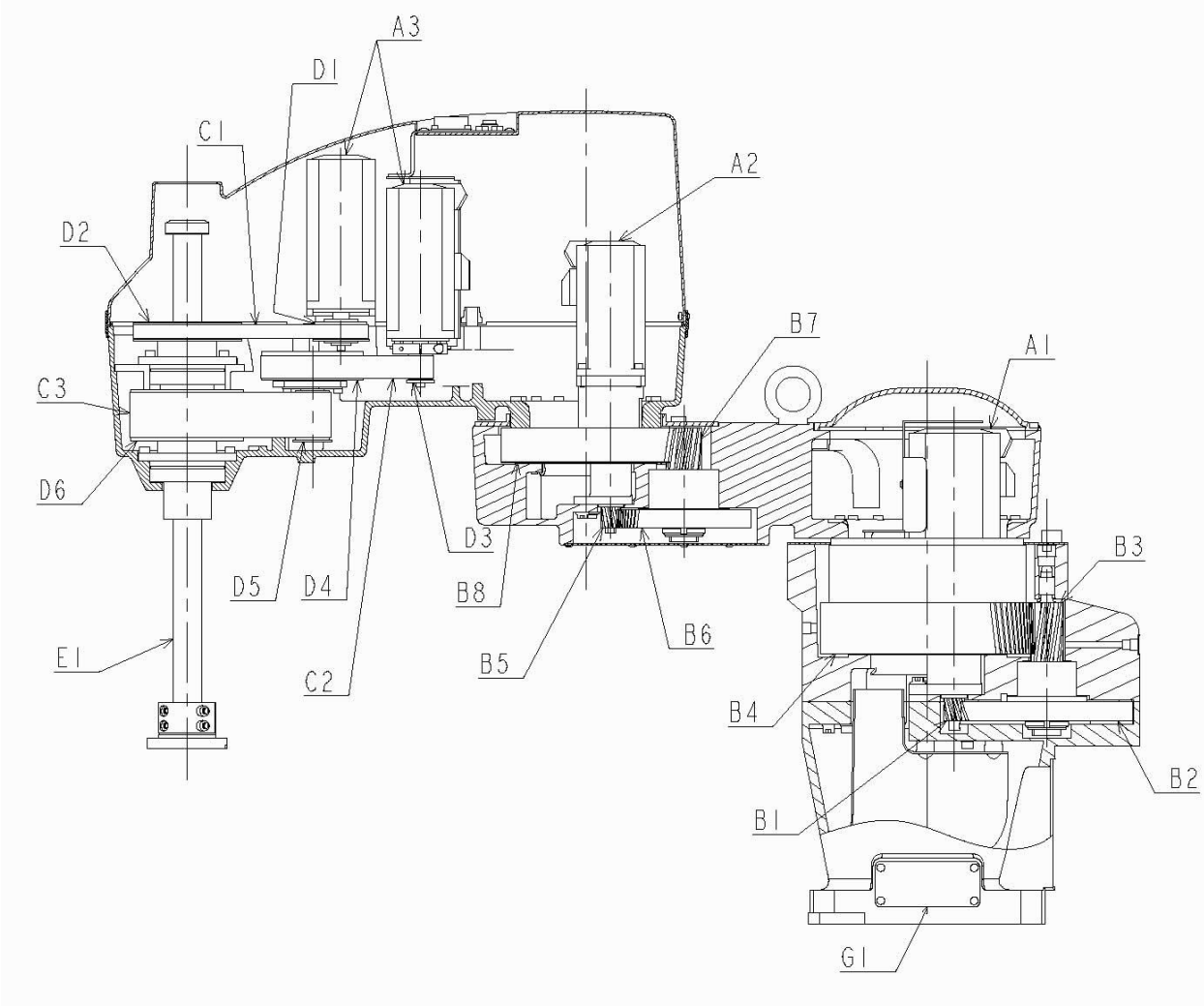
- With the exception of the grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts by the user.

Attachment figure A Locations of the replacement parts for maintenance

Locations of the TH450/TH550/TH450-T/Th550-T replacement parts for maintenance
The model of the figure below is TH450. Though there are a few differences by the difference between the model and the specification, basic locations are shown in the figure below.



Locations of the TH650/TH850/TH1050 replacement parts for maintenance
The model of the figure below is TH650. Though there are a few differences by the difference between the model and the specification, basic locations are shown in the figure below. Harness is located in the inside of the robot.



Attachment 2List of Controller Replacement Parts for Maintenance

Part name		Type	Unit Code	Maker	Q'ty	Remarks
PS1 (Switch power supply)		LEB100F-0524		Cosel	1	
PS2 (Switch power supply) (Only TS2100)		LDA15F-24		Cosel	1	
Main power switches	TS2000	IR-11-AB-10A-1B		Nippon Thermo	1	
	TS2100	ICP25-L-111-108-10		Nippon Thermo	1	
Lithium battery		ER3V-C		Toshiba Denchi	1	
Fuse (X8HN and X8HI printed boards)		51NM030H		PICO	1	
Printed circuit boards	Common	X8HC (Main Control)	Y610A2100	SHIBAURA MACHINE CO.,LTD.	1	
		X8HJ (Connector)	Y610A2320		1	
		X8HN (I/O printed board)	Y610A2120		1	Minus (-) common
		X8HI (I/O printed board)	Y610A2130		1	Plus (+) common
	TS2000	X8HS (Servo)	Attachment 2.1			2 axis per board
		X8HP (Servo power supply)	Y610A2340		1	
		X8HBA (Mother) X8HBB (Mother)	Y610A2140		1 1	Standard: For 4-axis control Special: 5-axis control, etc.
	TS2100	X8HW (Servo)	Attachment 2.1			1 axis per board
		X8HH (Servo power supply)	Y610A2190		1	
		X8HM (Mother)	Y610A21A0		1	Standard: For 5-axis control
Operation panel unit		X8HO	Y610A2310		1	
Teach pendant		TP1000	Y610A2600		1	With 5m-long cable
System disk		TS2000SYS			1	

Attachment 2.1 TH Series Servo Amplifier Application table

Controller	Robot model	Servo Amp. Module Type	Unit Code	Q'ty	Remarks
TS2000	TH450/TH550/ TH450-T/TH550-T	X8HS 10A/30A	Y610A23L0	1	Axis 1, axis 4
		X8HS 10A/15A	Y610A23E0	1	Axis 2, axis 3
TS2100	TH650	X8HW 50A	Y610A2180	1	Axis 1
		X8HS 20A Single axis	Y610A23Y0	1	Axis 2
		X8HS 15A/15A	Y610A23H0	1	Axis 3, axis 4
	TH850/TH1050	X8HW 50A	Y610A2180	1	Axis 1
		X8HW 25A	Y610A21L0	1	Axis 2
		X8HS 20A Single axis	Y610A23Y0	2	Axis 3, axis 4
TS2100	TH650	X8HS 05A/20A	Y610A23D0	1	Axis 2(3)
	TH850/TH1050	X8HS 20A/20A	Y610A23A0	2	Axis 3, axis 4(3)

(3) : Other axis are same as the above table of TS2100.

Appendix Diagram 1 Component Connection Diagram

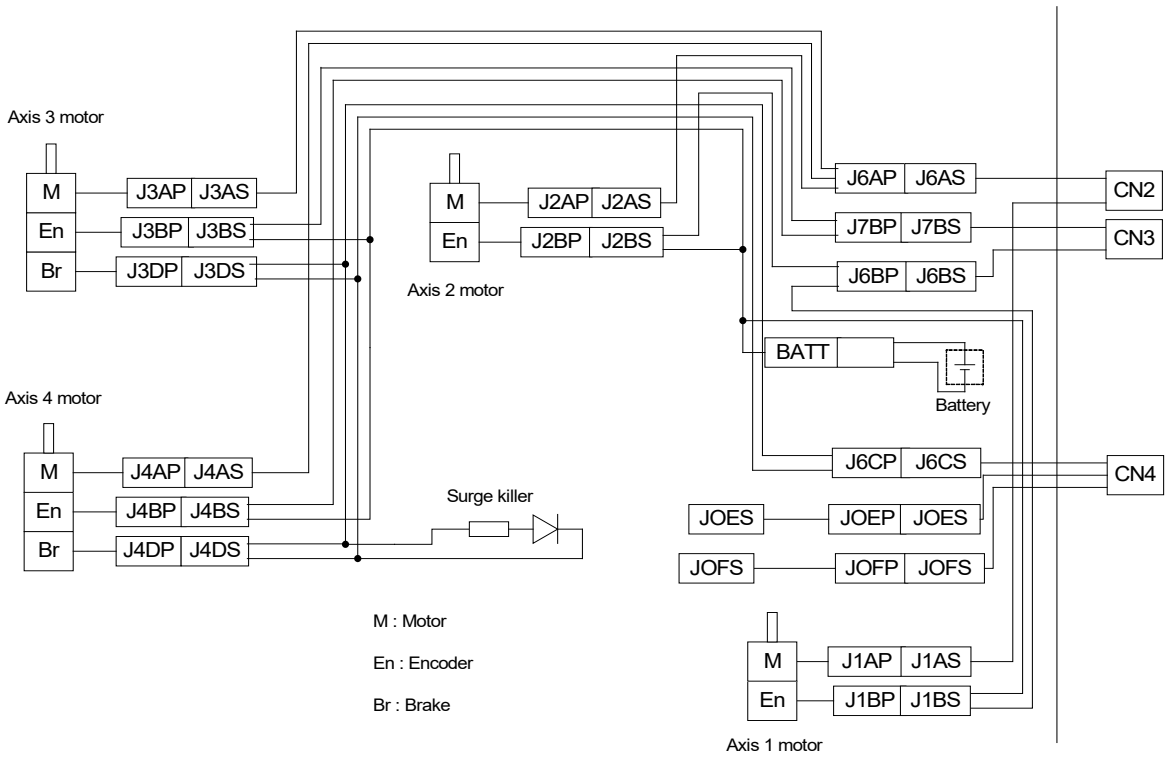


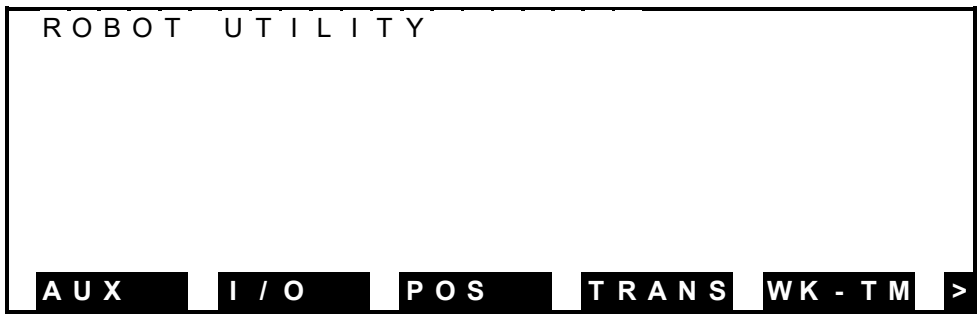
Fig. 4.1 Component connection diagram

Appendix A Encoder Error Reset

If the battery error or position detector error occurred and it will occur repeatedly even after the cause has been removed, reset the error flag (Err-df) in the encoder according to the following procedures.

[Procedures]

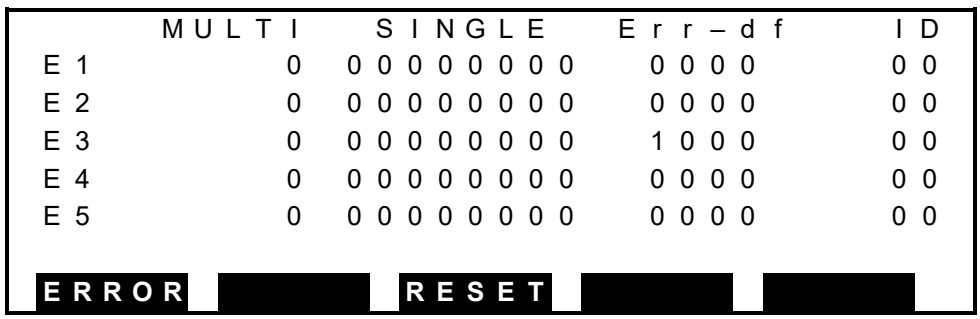
1. Press the UTILITY key, and the following screen appears.



2. Press the NEXT key.



3. Press the [ENC] key.



The encoder data of each axis is displayed.
Move the cursor to the position where the error flag is set to "1", using the ⇩ and ⇧ keys. Then enter "R E S E T" by means of the alphabet keys, and the error flag is reset compulsively. If the error still remains even after this operation has been done, it is yet to be cancelled. When this happens, investigate into the cause again.

Even if the error has been reset, make sure of the robot home position before operating the robot.

Appendix B Restoring Home Position Data

Batteries which back up the positional information of the position detectors (i.e., encoders) are contained in the robot base unit. Unless the robot is used for a long period of time, the battery voltage drops and the home position data will be lost. Also, if the voltage of the battery which backs up the internal memory of the X8HC printed board drops, programs, parameters and absolute coordinate data set by the user will be lost. To avoid this, replace the batteries on a regular basis. The yardstick of each replacement period is shown below.

- Batteries for position detectors ⋯ Two and a half (2.5) years after delivery (when the operating time per day is eight (8) hours).
- Battery for X8HC printed board SRAM ⋯ Five (5) years after delivery

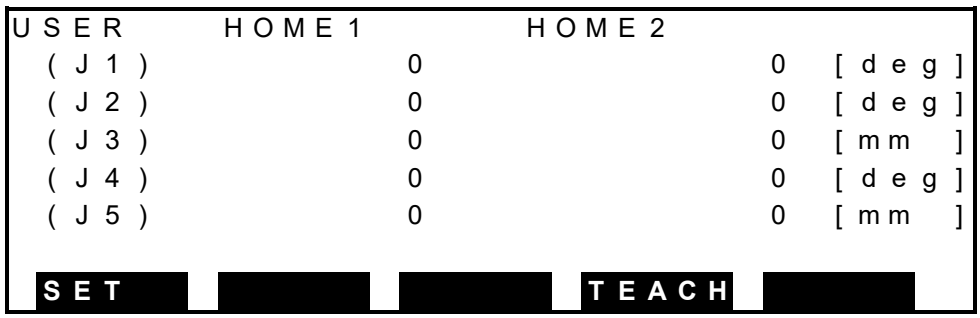
If the home position data has been destroyed due to battery voltage drop, etc., observe the following procedures to restore it.

a) How to restore the home positions

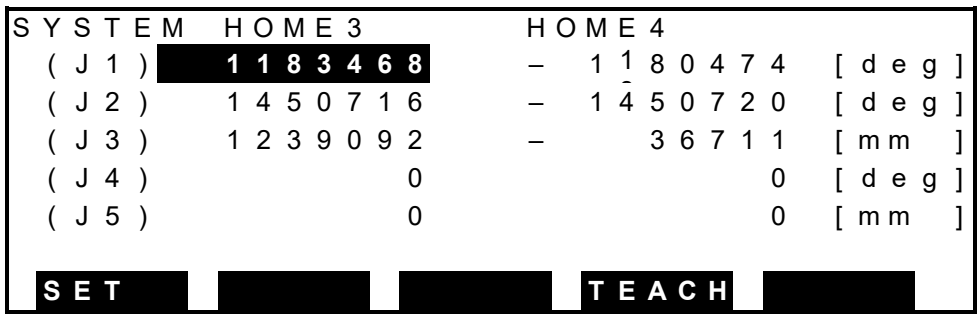
(1) Descriptions of HOME function

This function is useful to restore the original home position data by memorizing each axis coordinate data beforehand, if the machine home position data has been destroyed.
The set points can be assigned to HOME1 ~ 4. The plus (+) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME3, and the minus (–) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME4. The axis 4 is set at the 0° position according to the match-mark. In HOME1 and HOME2, the user can specify any desired positions. We recommend to set there the positions securing the robot end to prepare for the system construction.

[Page 1]



[Page 2]

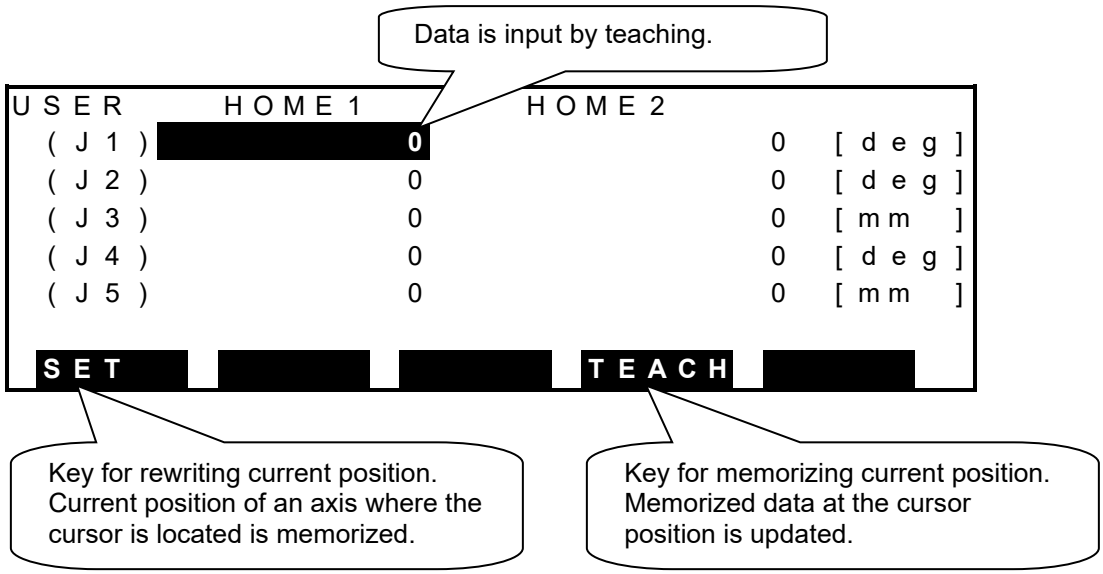


Note: The above are the screens for restoring the machine home positions. NEVER operate them at other than the restoration.

(2) Operating procedures

I. How to set the data in HOME1 and HOME2:
This paragraph describes the data setting in HOME1. (The same procedures are applied to HOME2.)

[1] Guide the robot to secure the tool shaft at a desired position.
[2] Turn off the servo power.
[3] Press the UTILITY key on the teach pendant.
[4] Press the NEXT key until [REORG] appears, then press the [REORG] key.



[5] When the above screen appears, memorize the position where the robot was secured in the [TEACH] mode, by moving the cursor to each axis.

II. How to restore the home position data by HOME1 (or HOME2):
If the machine home position coordinate data has been destroyed, the machine home position can be restored by means of the HOME1 function.

[1] Guide the robot to the position set by HOME1 and secure the robot there.
[2] Turn off the servo power.
[3] Press the UTILITY key on the teach pendant.
[4] Press the NEXT key until [REORG] appears, then press the [REORG] key.
[5] Move the cursor to each axis in the HOME1 mode of the REORG screen. Press the [SET] key and write the position where the robot was secured to the coordinate value set in HOME1. Thus, the machine home position can be restored.

Caution

Cautions on restoring home positions of axis 3 and axis 4 (REORG)

Our SCARA robot (TH series) will cause an axis interference between the axis 3 and axis 4. Unless you observe the predetermined procedures, the current position of the robot may shift. If you restore the axis 3 (or axis 4) home position, using the REORG function, be sure to restore the axis 4 (or axis 3) home position also.

<Operating procedures>

[1] Secure the axis 3 and axis 4 to the mechanical stoppers or specified positions.

[2] Set the cursor to (J3) and press the [SET] (F1) key to reproduce the axis 3 home point.

[3] Likewise, set the cursor to (J4) and press the [SET] (F1) key to reproduce the axis 4 home point.

III. How to restore the position data by HOME3 (or HOME4):

The plus (+) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME3, and the minus (–) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME4. The axis 4 is set at the 0° position according to the match-mark.

The machine home position can be restored, using these data.

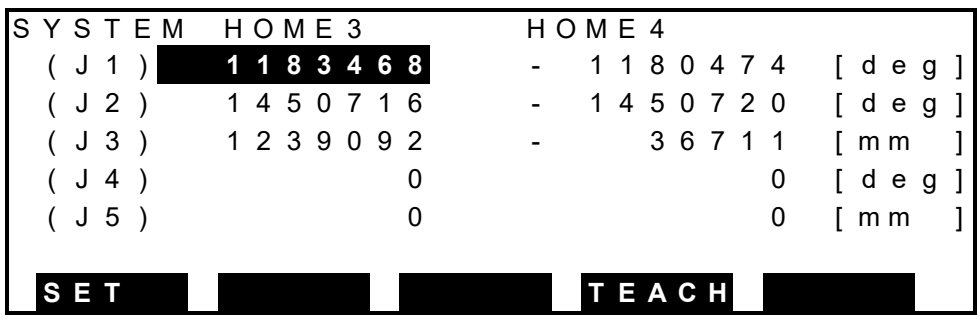
This paragraph deals with the restoration procedures, using HOME3.

[1] Turn off the servo power.

[2] Press the UTILITY key on the teach pendant.

[3] Press the NEXT key until [REORG] appears, then press the [REORG] key.

[4] Press the NEXT key to display Page 2.



[5] Contact the axis 1 of the robot to the plus (+) side mechanical stopper by hand.
[6] Move the cursor to (J1) of HOME3, then press the [SET] key. Now, the machine home point of axis 1 is restored.
[7] Likewise, contact the axis 2 of the robot to the plus (+) side mechanical stopper by hand.
[8] Move the cursor to (J2) of HOME3, then press the [SET] key. Now, the machine home point of axis 2 is restored.
[9] Press the [ESC] key to escape from the REORG screen once.
[10] Turn on the servo power, and set the axes 3 and 4 in the servo-free state.
[11] Set the axis 4 to the match-mark.
[12] Contact the axis 3 of the robot to the plus (+) side mechanical stopper by hand.
[13] Turn off the servo power.
[14] Repeat Steps [2] to [4] inclusive to call the HOME screen again.
[15] Move the cursor to (J4) of HOME3, then press the [SET] key. Now, the machine home point of axis 4 is restored.
[16] Move the cursor to (J3) of HOME3, then press the [SET] key. Now, the machine home point of axis 3 is restored.
[17] Press the [ESC] key to escape from the HOME screen. Now, the operation has finished.

Appendix C. Multi-turn data reset operation

Perform this operation if the mechanical connection between the servo motor and the mechanical unit is not changed, that is, for example, if the battery is replaced due to voltage drop or the cable is replaced due to breakage.
In the multi-turn data reset operation, set the axis to the home point match mark by hand and move it to the home position. A soft limit error may occur during the operation because the home position is outside the soft limit range. In this case, ignore the error until the operation is finished.
The multi-turn data reset operation resets only multi-turn data to zero without having any influence on data of a single turn or less. Thus, the operation can restore the data to the same status before the occurrence of the position encoder error.

! CAUTION

- If the robot is to be moved by hand in the power ON state, place it in the emergency stop state to assure safe operation.
- Be sure to assign two persons if the work is to be carried out while the axis-3 motor brake release switch is held down. One person carries out the work, and the other has the responsibility of observing the work while keeping a safe distance. The safety observer must turn off the controller immediately if an abnormality is detected. When the controller is turned off, the motor brake is activated even while the axis-3 motor brake release switch is held down.
- If the axis-3 motor brake release switch is pressed while the robot has a heavy load, the axis 3 may come down suddenly. Please keep your operators informed about this hazardous situation in order to prevent any accidents.

! CAUTION

- The multi-turn data reset operation is always protected because it exerts a significant effect on the current position of the robot if performed incorrectly. Do not release the protection of multi-turn data reset operation, except when the operation is to be performed.
- After multi-turn data reset operation, the home position may be shifted by a distance for a rotation of the motor. In this case, adjust the home position of the axis again, and perform the multi-turn data reset operation again.

Note:

- A soft limit error may occur during the multi-turn data reset operation. In this case, ignore the error until the operation is finished.

1) Display the encoder status screen on the teach pendant according to "Appendix A."



(Display if the system contains the controller TS2000/TS2100 of version earlier than 3.00)

2) Set each of the axes for which multi-turn data reset operation is to be performed, to the home point match mark by hand, and move it to the home position. A slight error is permitted.

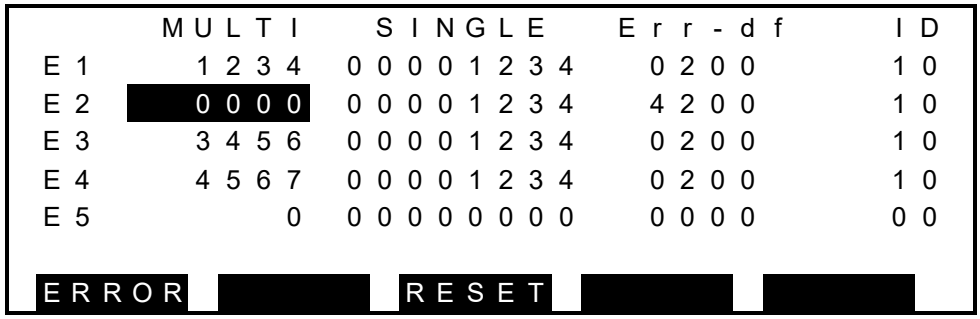
3) To release the protection, first press the [ALT] key and [0] key together.

4) The cursor can be moved after the protection is released. Move the cursor to the line of the axis for which multi-turn data reset operation is to be performed, using the cursor keys ([↑] and [↓] keys). The [←] and [→] keys are enabled if the system contains the controller TS2000/TS2100 of version 3.00 or later, and the cursor can be moved to the column MULTI. If the version is earlier than 3.00, the cursor keys are disabled and the cursor cannot be moved between columns.

5) For example, to reset the multi-turn data of axis 2, move the cursor to the cell where the line E2 meets the column MULTI.



6) Press [RESET] (F3 key), and then press the EXE key if there is no problem.



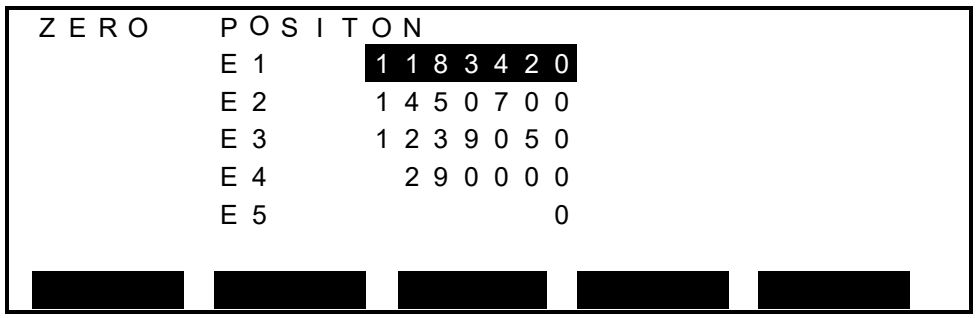
7) The multi-turn data of axis 2 is reset to zero by pressing the EXE key, and the encoder error is also reset at the same time.

8) Move to the general error screen, and press [RESET] (F3 key) to reset the error "8-129 Axis2 Encoder Abnormal."

9) To reflect the reset of multi-turn data to the current position data, turn off the controller, and then turn it on.

10) Press the UTILITY key on the teach pendant, and hold down the NEXT key until [ZEROP] appears in the function menu.

11) The following home point setting screen will be displayed. The position of each axis is set in units of $x10^{-4}$ [deg] or [mm]. For example, "E1 1183420" indicates that "the axis 1 is at the 118.3420-degree position."

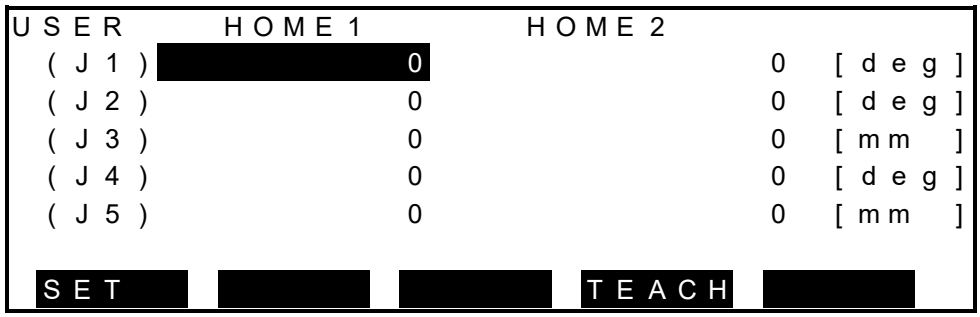


12) Record the values of E1 (axis 1) to E3 (axis 3) when each of the axes hits the mechanical stopper in the plus direction (or the minus direction) of the axis.

* For the axis 3 to hit the mechanical stopper, be sure to adjust the axis-4 home position to the home point match mark.

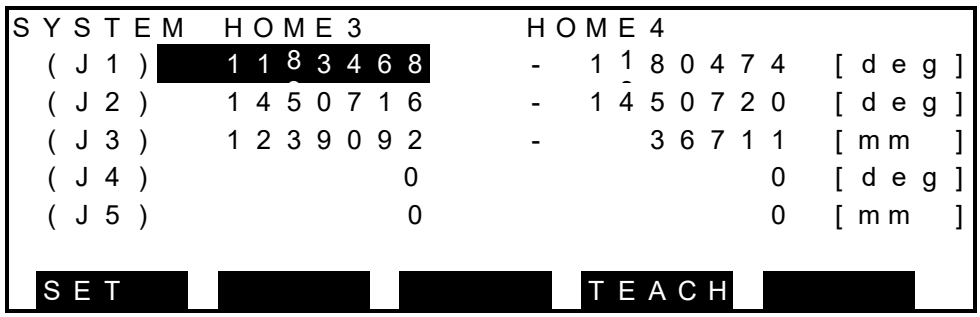
13) Press the [ESC] key on the teach pendant to return to the "ROBOT UTILITY" screen.

14) Hold down the [NEXT] key until [REORG] appears in the function menu, and press [REORG] ([F1] key).



15) Press the [NEXT] key to switch to the "SYSTEM HOME3 HOME4" screen.

* The plus (+) side mechanical stopper positions of axes 1 to 3 are factory-set in the column HOME3, and the minus (-) side mechanical stopper positions of axes 1 to 3 are factory-set in the column HOME4. There are no settings for the axis 4 because it has no mechanical stoppers.
"J1" is axis 1.



16) Compare the values of E1 (axis 1) to E3 (axis 3) recorded in step "12)" with the values of J1 to J3 in the column HOME3 (or HOME4). If the digits from the first to the third last digit of the recorded value of each axis are identical to those of the corresponding value in the column HOME3 (or HOME4), the data is restored to the same status before the occurrence of the position encoder error.
If the third last digit of the recorded value of each axis differs from that of the corresponding value in the column HOME3 (or HOME4), the axis cannot be returned to the home position correctly because of a slight shift generated when the axis was moved to the home position by setting to the home point match mark by hand. In this case, start the procedure from step "1)" again.

17) As for the axis 4, there are no mechanical stoppers, so the return to the home position cannot be checked through the teach pendant. To check if the axis is returned to the home position correctly, select the TEACHING mode and move the axis to the teach point at a low speed. If it is not returned to the home position correctly, start the procedure from step "1)" again.

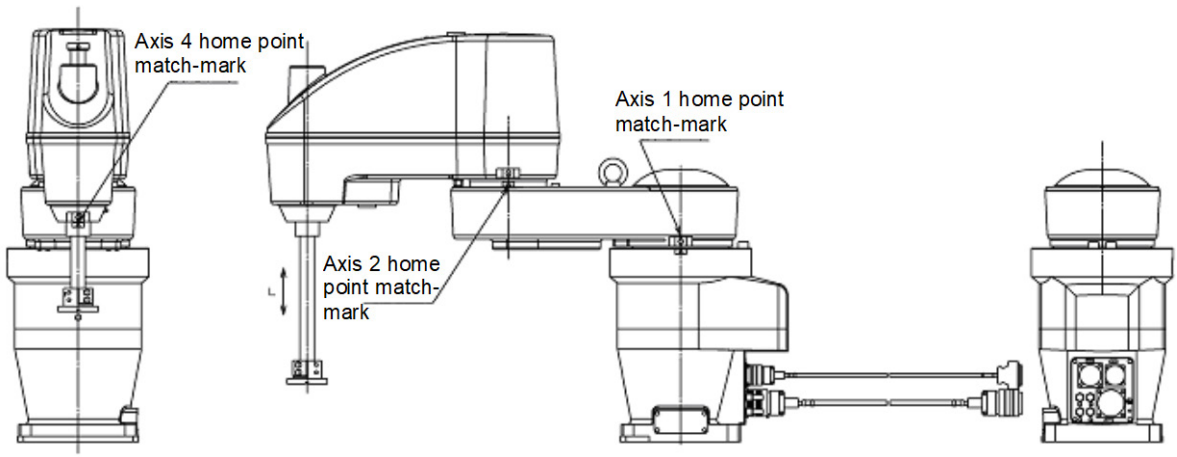
After multi-turn data reset operation, the home position may be shifted by a distance for a rotation of the motor.
The following table shows the shifts of each axis for a rotation of the motors TH450 to TH1050:

	Axis 1 [deg]	Axis 2 [deg]	Axis 3 [mm]	Axis 4 [deg]
TH450	7.200	7.200	30.000	27.356
TH550	4.500	7.200	30.000	27.356
TH650	4.497	7.126	25.000	25.821
TH850	3.747	5.496	25.000	15.652
TH1050	3.747	5.496	25.000	15.652

Appendix D Restoring Home Position Data

Batteries for backup of position information of the position detector are contained in the robot base. When the robot is not used for a long term, the battery voltage drops and the home position data will be lost accordingly. Also, when the battery for backup of SRAM of the X8C2 (MMI) printed board drops, absolute coordinate data set by the user will be lost. To avoid these, be sure to replace the batteries on a regular basis. (See the maintenance manual.)
If the robot arm is moved suddenly during power OFF, the absolute coordinate counter may not track the arm motion, resulting in destruction of the home position data.
If the home position data was destroyed by mistake, observe the following procedures to restore it.
This restoring method may cause a minor error according to the adjustment.

a) Turn on the controller power and align the home point match-marks of each axis in the manual mode. If this is not possible, mark the reference position of each axis and jot down its coordinate beforehand.



Locations of home point match-marks of respective axes

b) Move the axis 3 down to near the lower limit and set length "L" in the figure above (i.e., distance between the boss end surface of the tool shaft and tool flange end surface) to the following value.

TH450/TH550 : 155 mm
TH650 : 209 mm

c) Setting of new home position data (To be set through teach pendant.)

1) Select the TEACHING mode by means of the master mode switch.

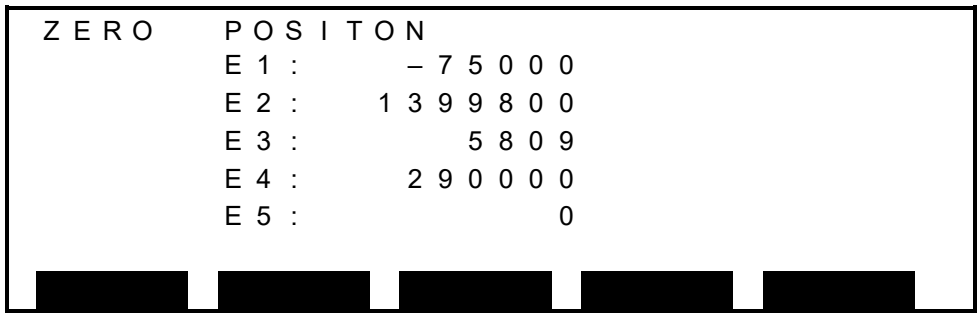
2) Press the UTILITY key on the teach pendant.

3) Press the NEXT key on the teach pendant twice.

4) Turn the servo power off.

5) Press the [F3] key on the teach pendant to select the ZEROP mode.

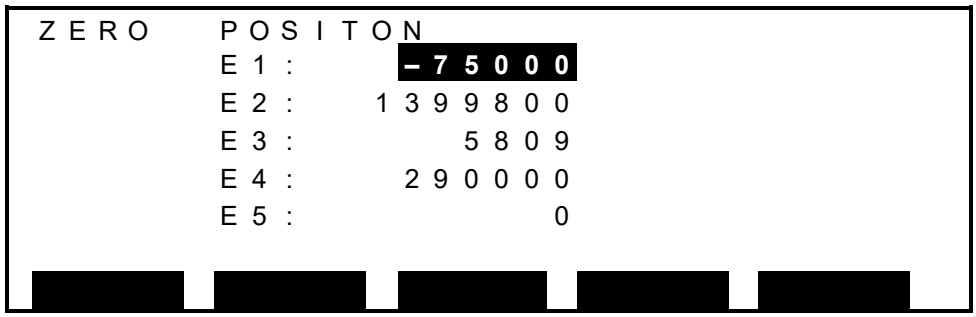
6) The following home point setting screen is called.



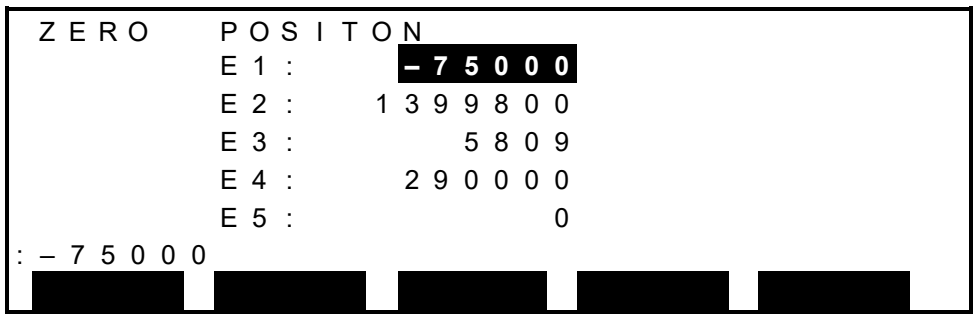
7) Press the ALT and 0 keys on the teach pendant.

8) The following home point editing screen appears.

9) The numerical value of E1 (axis 1) is inversely displayed.

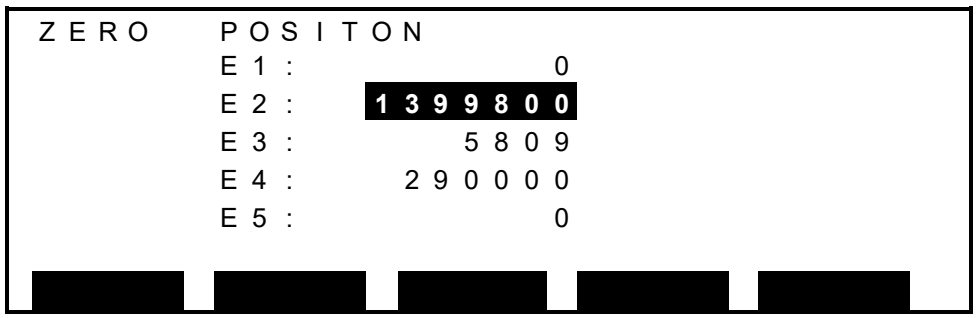


10) Press the EXE key on the teach pendant, and the value of "E1" is displayed at the lower left side. Set "0" in lieu of this value and press the EXE key. Then the E1 value becomes "0".
Unless the axis could be located at the home point mark, enter the coordinate value registered beforehand.



11) When the "↓" key on the teach pendant is pressed, the following E2 value is displayed by reversed image. Perform setting in the same manner as above.

Caution: Be sure to set the home point in the order of E1 → E2 → E4 → E3 because E3 (axis 3) and E4 (axis 4) of the SCARA robot are interlocked with each other.



d) Confirmation of home position

1) Make sure of the robot current position according to the POS data of the UTILITY mode. Guide the robot to the approximate home position by manual operation and confirm the position. Unless it could be located at the home point mark, make sure that the approximate position is the reference position marked in advance.

2) Execute the following program and make sure of the robot machine home position. Unless the robot could be located at the home point mark, DO NOT use the following program.
Create a program in which the robot moves to the position recorded beforehand, referring to the following program. Then make sure that the robot moves to the reference position mark according to the program thus created.

```
PROGRAM ZERO
 SPEED=30
 MOVEA 3,100
 MOVEA 1,0
 MOVEA 2,0
 MOVEA 3,0
 MOVEA 4,0
END
```

e) Data saving
Save the system parameters in the system disk.
Connect the personal computer and controller using TSPC (option), insert the system disk into the floppy disk unit and copy the "MACHINE.PAR" file from the robot controller to the personal computer.
Save the home position data in the system parameter file. The home position data is the data characteristic of the robot, which differs with the robot you use. If you effect a cold start without saving this data, you have to restore the home position data again. To avoid this, be sure to save the data.

f) Now, the operation completes.